

FALCON's Response to Decathlon's RFQ for 'Robo Shuttle Automation System' dated 27 Feb,24

March 5, 2025



www.falconautotech.com

Kind Attention –

Mr Jilson Martin

Offer Ref: F23-00439_Rev 5_Temp; Date 05-03-2025

Subject – Techno-Commercial Offer for NEO with Conveyor Automation

Dear Team,

We are pleased to submit our Techno-Commercial Offer to Decathlon in response to your RFQ for 'Robo Shuttle Automation System'. This particular proposal is for your Bangaluru , India warehouse and the solution is customized according to the specific needs and processes recorder by us in this warehouse.

We would like to highlight the fact that Falcon Autotech is a global leader in providing intra-logistics automation solutions. Falcon has installed over **150 sorters worldwide** with capacity ranging from **800 to 60,000 PPH** and has an excellent track record in designing, manufacturing, and installing parcel sortation systems.

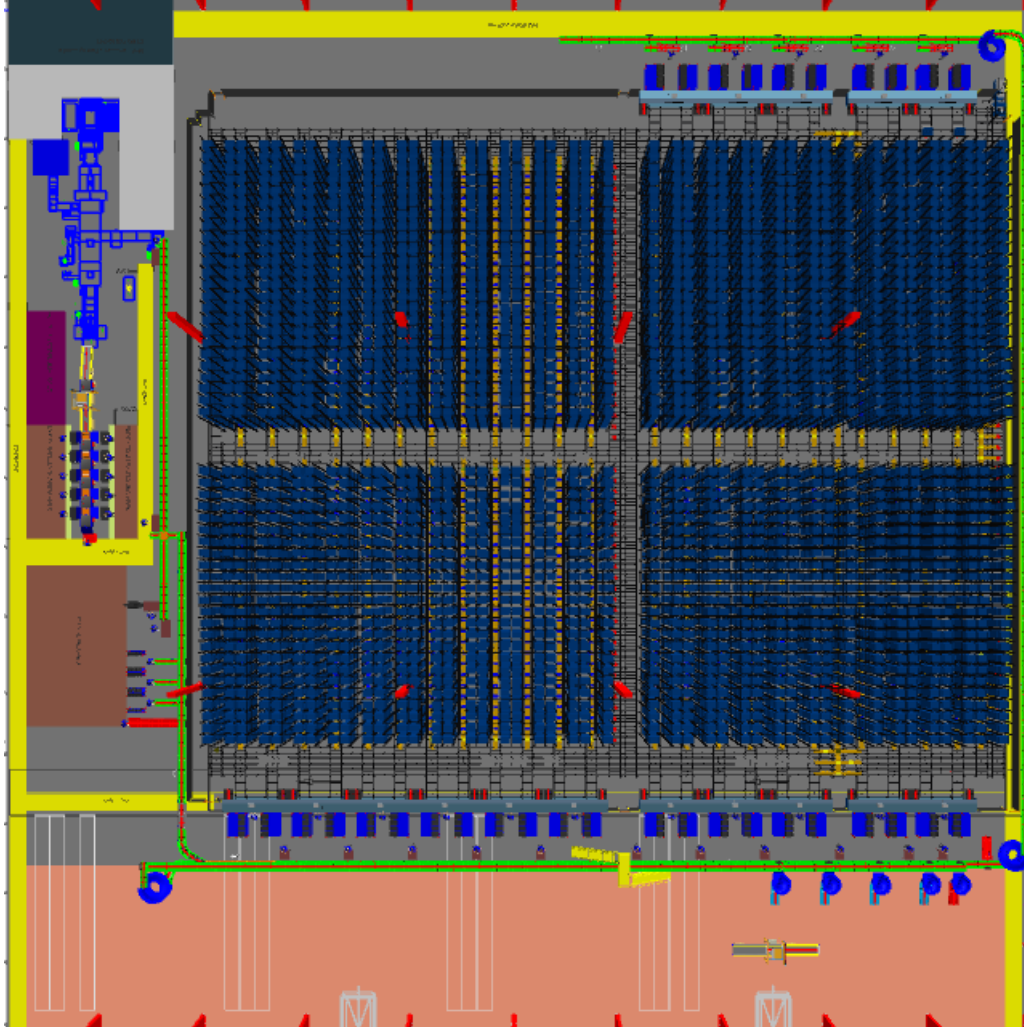
As you will note, we have studied your requirements in great depth and, along with the information collected during the workshops, visits and calls with your different team , we have put together a detailed technical proposal laid out in various sections and sequenced to enable you to understand our proposed solution and to re-enforce our commitment to be your partner in this strategic initiative.

To conclude, I would like to add my personal commitment, on behalf of Falcon Autotech. As we move through the RFP process, please do not hesitate to contact me and my team. We will be pleased to assist you with any further information or clarifications that you might have.

Best Regards,

Rajasik Sen

Response to Decathlon's RFQ for Robo Shuttle Automation Solution



Contact Person	:	Rajasik Sen
Title	:	Sr Manager
Mobile	:	+91 9999720414
Email	:	Rajasik.sen@falconautoonline.com
Company Name	:	Falcon Autotech Pvt. Ltd.
Website	:	www.falconautotech.com
LinkedID	:	https://www.linkedin.com/company/falconautotech/mycompany/

Table of Contents

1.	Glossary	6
2.	Executive Summary	7
3.	Company Profile.....	8
4.	Falcon's Experience and Achievements in Sortation Space Globally	12
5.	Reference Projects.....	14
5.1	Project 1- (CEP Client, UK).....	14
5.2	Project 2- (CEP Client, Riyadh).....	15
5.3	Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)	16
5.4	Project 4- (CEP Client, Sydney).....	17
5.5	Project 5- (Client – E-Commerce, India)	18
5.6	Project 6- (Ecommerce Client, India)	19
6.	Handled Parcel Spectrum for Conveyor Setups Minimum Maximum	20
6.1	Carton & Bin size Conveyable on the entire system.....	20
6.2	Types of Parcels	20
7.	Proposed System Description/Description of Technical Equipment.....	21
7.1	Objective	21
7.2	Summary of the System.....	22
7.2.1	Neo ASRS System:	22
7.2.2	Solution Layout Description:	23
7.2.3	Storage Grid:	24
7.2.4	Put-Away stations & Conveyors:.....	25
8.	Process Flow.....	30
9.	Bot & Station Calculations.....	31
9.1	NEO.....	32
9.2	Powered Roller Conveyor	35
9.3	Pop- Up Unit.....	36
9.4	Swivel Wheel Diverts	37
9.5	Pop-up Sorter for Ecommerce Parcels	38
9.6	Pick/Put to Light Module	39
9.7	Automatic Barcode Scanners.....	40
9.8	Hand-Held Barcode Scanners	40
9.9	NEO Bins.....	41

10.	Proposed System Technical Details/BOM	42
10.1	Mechanical equipment	42
10.1.1	Conveyor BOM	43
10.2	Electrical Equipment	45
10.3	Control System	45
11.	Falcon's WCS CONTROLIT	46
12.	Key Components Make	47
13.	Automation Management System (AMS)	48
13.1	Issue Logging	48
13.2	Preventive Maintenance Check List	49
13.3	Predictive Maintenance	49
13.4	Training Portal	50
13.5	Operational Analysis	50
14.	Program Organisation	51
14.1	Program/ Project Schedule	51
	 Error! Bookmark not defined.	
14.2	Program Management	51
15.	Commercial Terms	52
15.1	Price Sheet – Consolidated	52
15.2	Payment Terms	52
16.	Warranty & SLA	53
16.1	Supply and Service Terms	53
17.	Exclusions	60
18.	Annexure*	60

1. Glossary

S. No.	Term	Description
1	RFQ	Request For Quote
2	GTP	Goods to person
3	PPH	Parcels Per Hour
4	AC	Alternating Current
5	DC	Direct Current
6	PLC	Programmable Logic Controller
7	IT	Information Technology
8	BOQ	Bill Of Quantity
9	I/O	Input/ Output
10	PDP	Power Distribution Panel
11	PC	Personal Computer
12	UPS	Uninterrupted Power Supply
13	PTL	Pick/Put to Light

2. Executive Summary

Falcon is pleased to confirm its interest in responding to the automation requirement presented by Decathlon. Our team has been working closely with the relevant stakeholders, with a clear commitment to listening to and understanding your needs and ensuring this project's success.

As prime contractor, Falcon ensures its full commitment to successfully completing this project. Following the same objective for the said system, we are happy to offer a compliant solution meeting all technical and operational requirements, high-performance, optimized, tailor-made, fast and secure planning, and a competitive price.

Our solution is based on the following key characteristics:

- **NEO ASRS system with 41812 bin positions**
- **16 GTP stations with PTL's for order consolidation as well as smart put-away.**
- **Automatic movement of put away material from drop location to put-stations.**
- **Automated movement of packed orders and empty totes through independent conveyor systems.**
- **Automated sorting line for packed orders.**
- **Dedicated conveyor system for Ecommerce orders with multi-consolidation zone with PTL's**
- **Automatic sorting line for ecommerce packages for courier partner-wise sorting.**

A tailor-made and simple layout, specifically designed for Decathlon. The proposed layout is the result of the technical requirements listed during our various meetings and E-mail exchanges.

Note : Layout and numbers defined above may change at detailed engineering stage due to incorporation of in-rack sprinklers etc.

3. Company Profile

Falcon Autotech (Falcon) is a global intralogistics automation solutions company. With over 10 years of experience, Falcon has worked with some of the most innovative brands in E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical Industries. With our proprietary software and robust hardware integration capabilities, Falcon designs, manufactures, supplies, implements, and maintains world-class warehouse automation systems globally. Falcon's strong research and development team and the continuous focus on innovation reflect our strong solution line around Sortation, Robotics, Conveying, Vision Systems and IOT. Falcon has done over 1,800 installations across 15 countries on four continents.

Falcon 2.0

Operational since in 2012, Falcon is a global intra-logistic automation solution provider handling parcels, bags, cartons and pieces

Falcon has five key solution lines: 3D Robotics, Sortation systems, Dimensions & Weight Systems, Put/pick To Light & Conveyors, along with rapidly scaling proprietary software called FACTS

Work with leading E-Commerce, CEP, Fashion, Food/FMCG, Auto and Pharmaceutical brands

5 Product Lines

15+ Countries with Live Installations

1800+ Total Installed Systems Globally

1Million+ Sorts Per Hour

600+ Employees

Long-standing Relationships with Industry leaders

Ecom Retail

Flipkart

amazon

Lazada

noon

CEP

aramex

BLUE DART

DELHIVERY

Ecom Express

Distribution & Others

G

SWIGGY

TITAN

Hindustan Unilever Limited

News & Articles

FALCON AUTOTECH EXPANDS PRESENCE IN THE MIDDLE EAST WITH NEW OFFICE OPENING
Falcon Autotech is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market.

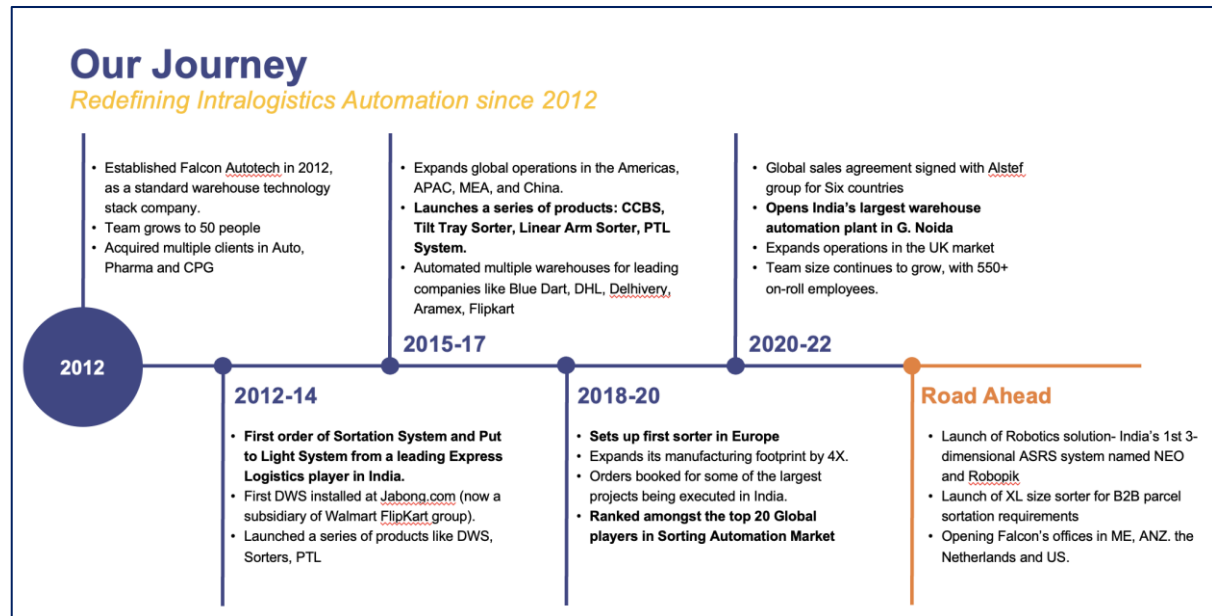
FALCON AND ALSTEF GROUP ANNOUNCE GLOBAL TECHNOLOGY PARTNERSHIP
Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions
Published - Sept 2022

Falcon Autotech is currently among the top 15 intralogistics automation companies; our vision is to become the top 10 intralogistics automation company in our focused product lines.

Our Vision

To be amongst the Top 10 global intra-logistics automation companies in our focused product lines

The team started out in 2004 solving special purpose automation problems for clients and later established Falcon Autotech in 2012 with strong focus on building standard technology stack spanning across Hardware, Firmware and Software to tackle bigger Supply Chain problems around warehouse automation and material handling. Over the decade, Falcon has made rapid strides and has carved out a niche in some of the world's most cutting-edge technologies: Sortation, Robotics, Conveying, Vision Systems and IOT.



As a leading player in the intra-logistics automation space, Falcon continuously strives to improve the operational efficiencies and accuracies for its clients through its domain knowledge and experience in addition to its wide range of products and solutions. To be able to live up to the high expectations set forth by our clients, the team at Falcon realizes the importance of taking up selective applications in focused Industries and delivering world-class projects in return.



Product and Solutions

With 100% focused on Parcels, Eatches, Totes, Bags, Cartons

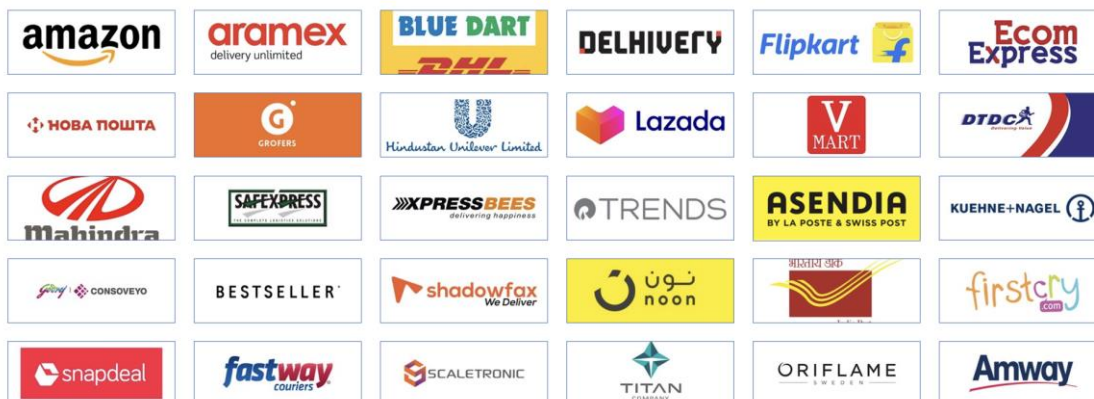


Falcon Autotech has successfully delivered warehouse automation solutions based on smart and innovative combinations of the above product lines for effective materials handling, sortation, and movement. The process is controlled in real-time by our in-house WCS applications. These solutions considerably cut the need for manual operations, improve working conditions, and ensure the highest accuracy of the entire process up to final delivery to the recipient.

Over the last 10 years, Falcon has worked with some of the most innovative brands worldwide and has established long-standing partnerships. These brands are testimony of our strong focus on delivering superior customer satisfaction and offering end-to-end intralogistics solutions.

Select Key Clients

Some of world's most innovative brands trust us with their intra-logistics warehouse automation requirements

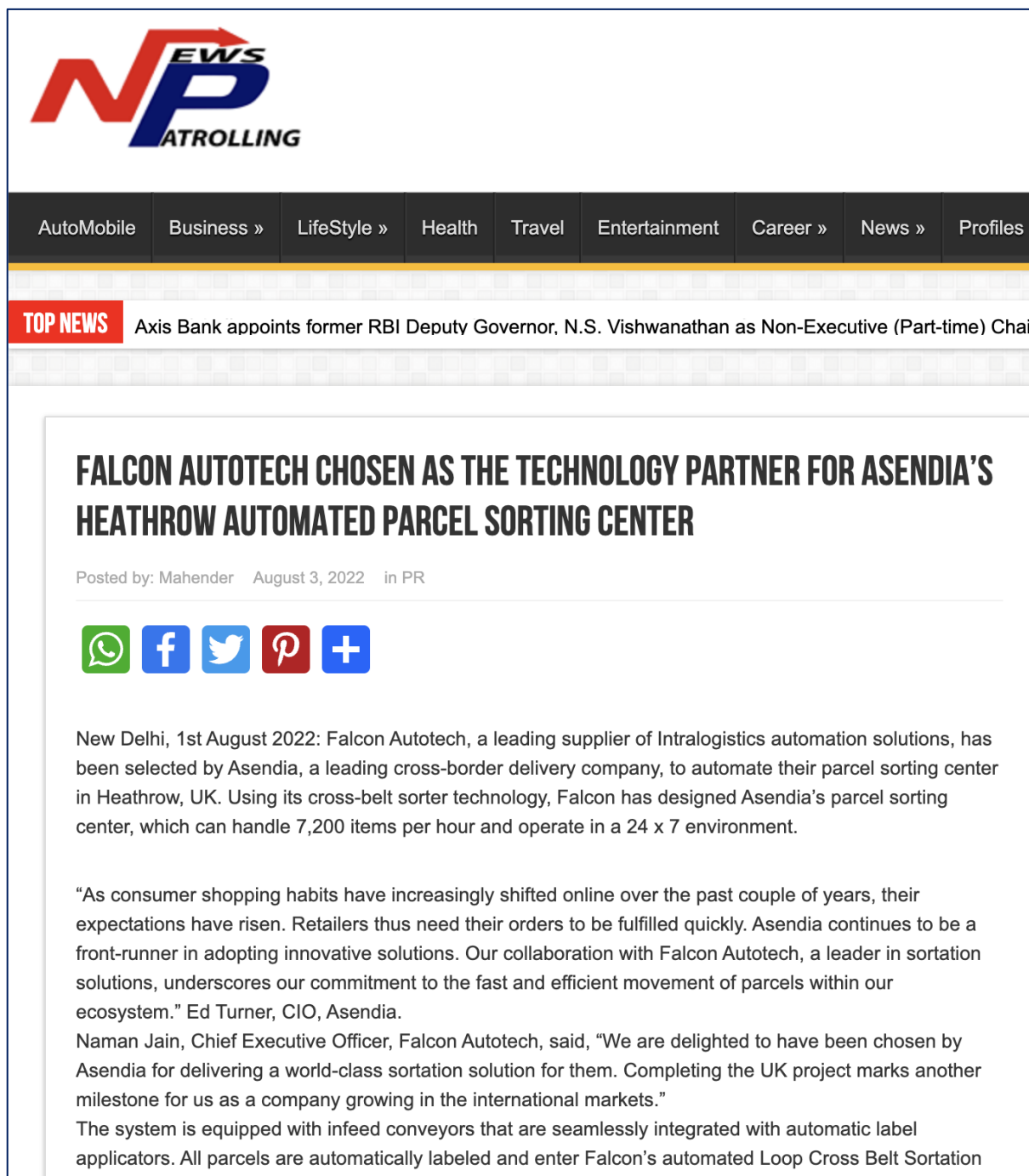


With over 1,800 installations, today Falcon's systems are used all over the globe. Falcon has highly motivated team of 600+ employees supported by over 15 global partners who help us design, manufacture, deliver and maintain automation solutions globally.



4. Falcon's Experience and Achievements in Sortation Space Globally

- Ranked among **Top 20 Sortation System Suppliers globally**. Only company from APAC/EMEA Region.
- Currently possess one of **the World's largest portfolios in Sortation Technologies**: 7 In-house technologies.
- Total installed capacity of **10 million Parcels per day** worldwide.
- Only company to be able to offer a **Fully Integrated AMS**.



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TOP NEWS Axis Bank appoints former RBI Deputy Governor, N.S. Vishwanathan as Non-Executive (Part-time) Chair

FALCON AUTOTECH CHOSEN AS THE TECHNOLOGY PARTNER FOR ASENDIA'S HEATHROW AUTOMATED PARCEL SORTING CENTER

Posted by: Mahender August 3, 2022 in PR

New Delhi, 1st August 2022: Falcon Autotech, a leading supplier of Intralogistics automation solutions, has been selected by Asendia, a leading cross-border delivery company, to automate their parcel sorting center in Heathrow, UK. Using its cross-belt sorter technology, Falcon has designed Asendia's parcel sorting center, which can handle 7,200 items per hour and operate in a 24 x 7 environment.

"As consumer shopping habits have increasingly shifted online over the past couple of years, their expectations have risen. Retailers thus need their orders to be fulfilled quickly. Asendia continues to be a front-runner in adopting innovative solutions. Our collaboration with Falcon Autotech, a leader in sortation solutions, underscores our commitment to the fast and efficient movement of parcels within our ecosystem." Ed Turner, CIO, Asendia.

Naman Jain, Chief Executive Officer, Falcon Autotech, said, "We are delighted to have been chosen by Asendia for delivering a world-class sortation solution for them. Completing the UK project marks another milestone for us as a company growing in the international markets."

The system is equipped with infeed conveyors that are seamlessly integrated with automatic label applicators. All parcels are automatically labeled and enter Falcon's automated Loop Cross Belt Sortation

POSTED ON APRIL 17, 2023 IN / NEWS

Falcon Autotech Expands Presence in the Middle East with New Office Opening

Dubai, UAE – 17th April 2023 – Falcon Autotech (Falcon) is pleased to announce the opening of its new office in Dubai, UAE, marking a significant milestone in the company's expansion into the Middle East market. The new location will enable Falcon to better serve its existing customers in the region and expand its reach to new markets and prospects.

The decision to establish a presence in the Middle East was driven by the growing demand for Falcon's solution in the region, as well as the strategic importance of the Middle East as a hub for business and innovation. With a team of experienced professionals based in the region,

Falcon Autotech and Alstef Group Announce Global Technology Partnership for Parcel Sortation Solutions

Falcon Autotech, a leading supplier of Intralogistics automation solutions, and Alstef Group, an expert in comprehensive baggage handling solutions and parcel automation integration announce a strategic technology partnership for parcel sorting solutions. As part of an exclusive distribution agreement, Alstef Group will exclusively expand the deployment of Falcon Autotech's Cross-belt sorter range to specific geographies. Falcon Autotech's Cross Belt Sorter (CBS) range includes a linear CBS and a loop CBS option. With proven features, such as automatic item centering, flexible chute positioning, wireless data communication, real-time video coding, and variable electric discharge, the Falcon CBS range is one of the fastest and smoothest sortation systems available in the market.

5. Reference Projects

Falcon has a strong legacy in **Warehousing Automation** solutions and references-

1. Expertise in Parcel Sortation, Piece Picking and Handling, Case Picking and Handling.
2. Lifecycle services (maintenance, spares supply chain, support).
3. Full **in-house** expertise (Hardware/Software).
4. Turn-key **tailored** solutions.

The references list presented below focusing on Sortation Solution –

5.1 Project 1- (CEP Client, UK)

This Solution is designed to handle a volume of 7200 parcels per hour. The system is equipped with three infeed conveyors integrated with automatic label applicator before parcels enter the sortation system. The parcels are sorted using Falcon's Loop Cross Belt Sorter equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities. The sorter is installed on the mezzanine floor and sorts directly to 58 end destinations.

Solution Specifications –

- Throughput: 7200 PPH
- End Destinations: 58 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.2 Project 2- (CEP Client, Riyadh)

In 2019, customers selected Falcon Autotech as a preferred supplier for its airport hub to supply linear cross belt sorter for processing parcels arriving via Air from different states and countries to distribute them locally. The customer chose Falcon Autotech based on its strong track record of success in providing intralogistics automation solutions, optimized design, software integration capabilities and life cycle support services.

Solution Specifications –

- Throughput: 4800 PPH
- End Destinations: 52 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- ARB Conveyor.
- Automatic Label Applicators.
- Linear Cross Belt Sorter.
- Specialized Chutes for Gentle Parcel handling.
- FOCR Engine.
- WCS Software System.

Site Picture –



5.3 Project 3- (Client – Courier Express Parcel, Dubai & Jeddah)

This Solution is designed to handle the bulk volumes through Launchpads and induct them into Falcon's Linear Arm Sorter. This system is designed for a throughput of 3000 parcels per hour equipped with automatic barcode scanning, dimensioning, weighing and image capture capabilities to into a total of 480 end destinations with a combination of primary and secondary sortation system. The secondary sortation is achieved by integrating put to light system.

Solution Specifications –

- Throughput: 3000 PPH
- End Destinations: 480 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- Linear arm sorter.
- Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- WCS Software System.

Site Picture –



5.4 Project 4- (CEP Client, Sydney)

The solution is designed for handling a throughput of 16,000 parcels per hour with the help of Falcon's Loop Cross Belt Sorter. The system consists of 2 feeding zones with a total of 10 feedlines. Sorter design enables the van drivers to directly drop the parcels at the dock doors. It has a total of 369 end destinations that are achieved with a combination of direct drops and PTLs. The system is integrated with 5 side automatic barcode scanning, weight and volume measurement and automatic detection of oversize and overweight parcels.

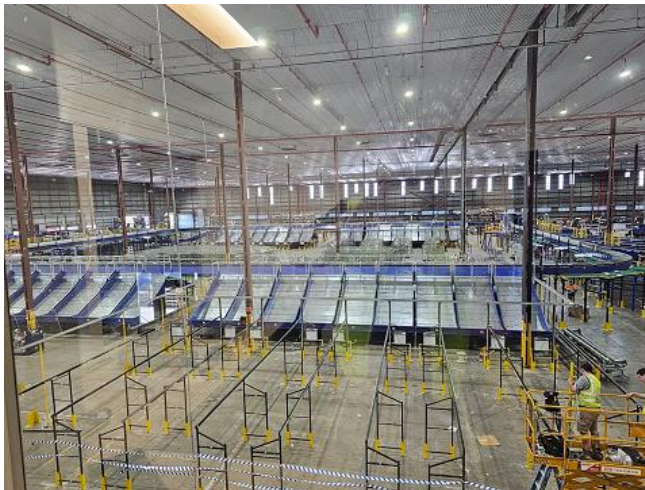
Solution Specifications –

- Throughput: 16000 PPH
- End Destinations: 369 Nos

Key Technology Modules –

- Powered Belt Conveyors.
- 2 Induct zone.
- 5 side Automatic Barcode Scanner.
- Automatic Weight & Volume Measurement System.
- Automatic detection of oversize parcel.
- Loop Cross Belt Sorter.
- WCS Software System.

Site Picture –



5.5 Project 5- (Client – E-Commerce, India)

Use Case – Destination Sorting of Packed Parcels.

In 2019, Client was looking for a potential automation partner for design and development of a new automated sortation system for B2C parcels. The system should be able to provide maximum uptime with reduced dependency on skilled manpower, and space optimization.

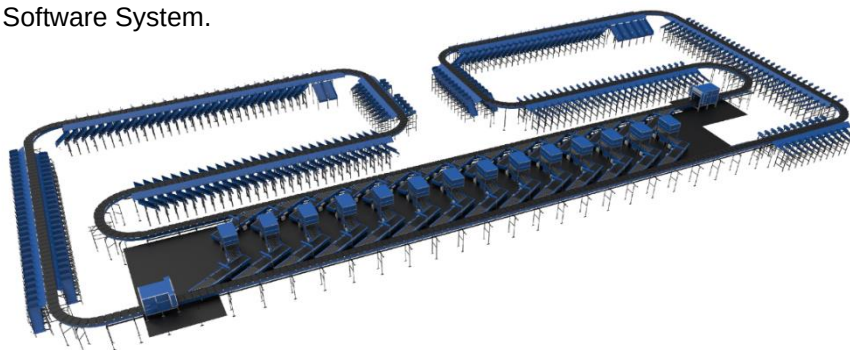
The customer chose Falcon Autotech based on its unique design which could cater to all their pain points, capability of seamless integration with WMS and life cycle support services.

Solution Specifications –

- Throughput: 27,600 PPH
- End Destinations: 410 Direct Outputs
- Building Size: 200,000 Sq. Ft

Key Technology Modules –

- Bulk Infeed Conveyors.
- ARB based Volume Distribution System
- Integrated Pre-sort System.
- Irregular Ejection System.
- Automatic Induct Lines.
- Automatic Barcode Scanner with Image Capture.
- Automatic Weight & Volume Measurement System.
- Loop Cross Belt Sorter.
- Smart Sliding Chutes for Direct bagging and Cage Sorting.
- Bag take out system.
- WCS Software System.



5.6 Project 6- (Ecommerce Client, India)

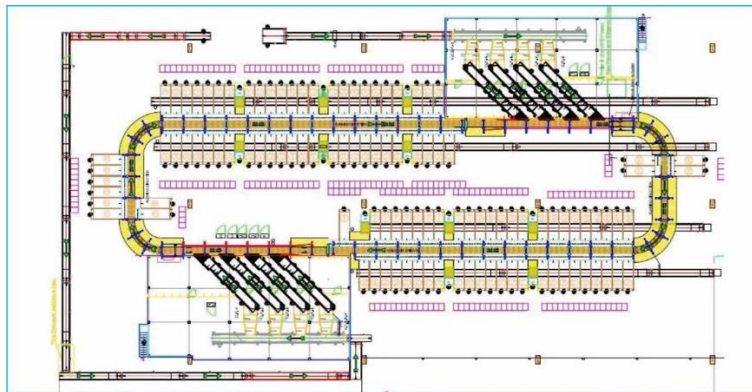
Use Case - Destination Sorting of Packed Shipments

Solution Specifications –

- Throughput: 24,000 PPH
- End Destinations: 110
- Building Size: 200,000 Sq. Ft

Key Technology Modules - Loop Cross Belt Sorter with a cell size of 900 x 500 mm

Layout and Site Pictures -



6. Handled Parcel Spectrum for Conveyor Setup

6.1 Carton & Bin size Conveyable on the entire system.

Max Length	mm	600
Max Width	mm	400
Max Height	mm	400
Max Weight	Kg	25
Min length	mm	250
Min Width	mm	100
Min Height	mm	50
Min Weight	Kg	5

6.2 Types of Parcels

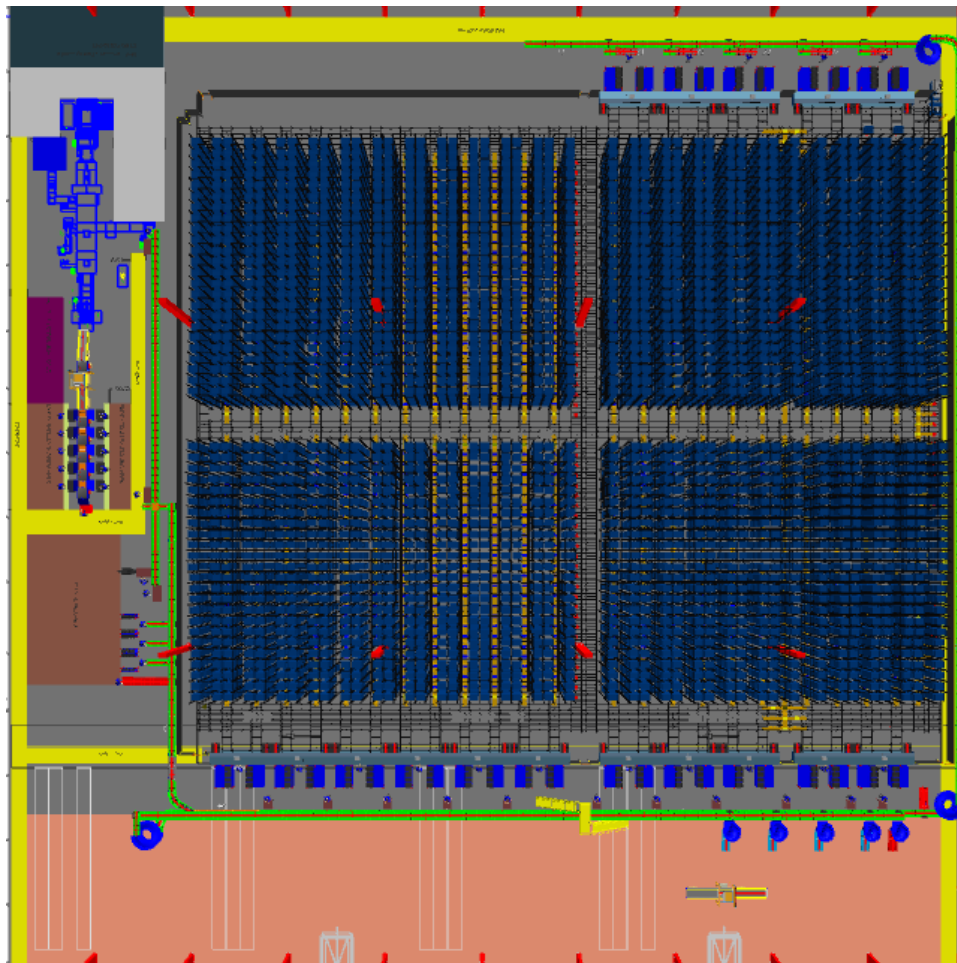
Plastic Totes and Corrugated Boxes for conveyors & PTL's	Min: L-250 X W-100 X H-50 MM Max: L-600 X W-400 X H-400 MM

7. Proposed System Description/Description of Technical Equipment

7.1 Objective

The purpose of this proposal is to present the design, manufacturing, installation, commissioning, testing, integration with Decathlon's WMS and Falcon's WCS and acceptance testing of the complete automation system comprising of the NEO -ASRS system and conveyor network to handle various up-stream and down-stream operations.

Proposed System Layout



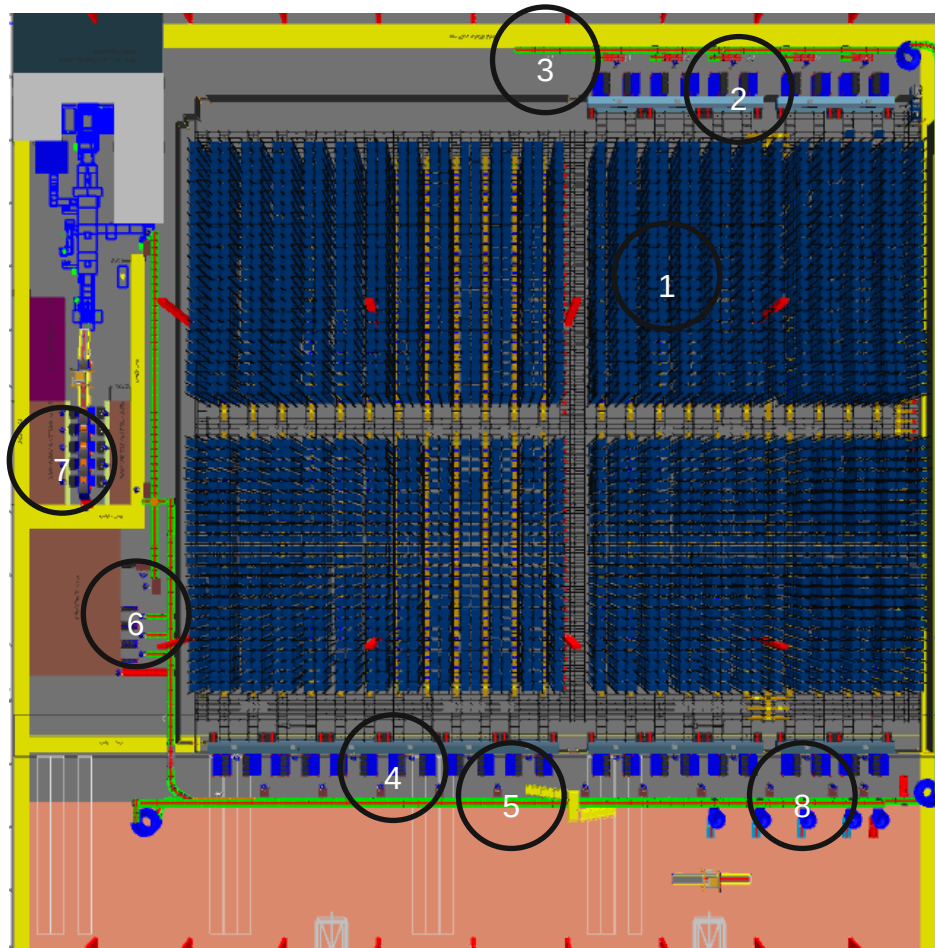
7.2 Summary of the System

7.2.1 Neo ASRS System:

Particular	Value
Total Storage (Bins)	41,812
Number of Rack Levels- Variable	22,21,20,19
Number of GTP stations	16
Number of Robots (14 hrs operation)	85
Number of PTL's in each station	24
Total Footprint of the storage grid and Bot tracks	~ 4200 sq. meter
Height of the storage Grid – Max/Min	12.05/10.05
Total Conveyor system length	~ 360 meters
Store Order sorting line – No of sorting splits	5
Ecommerce consolidation with PTL's	3 zones
Ecommerce package sorting line – No of sorting splits	10

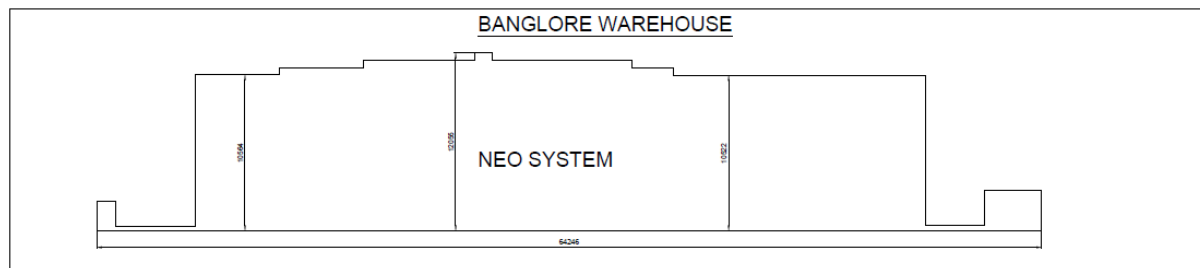
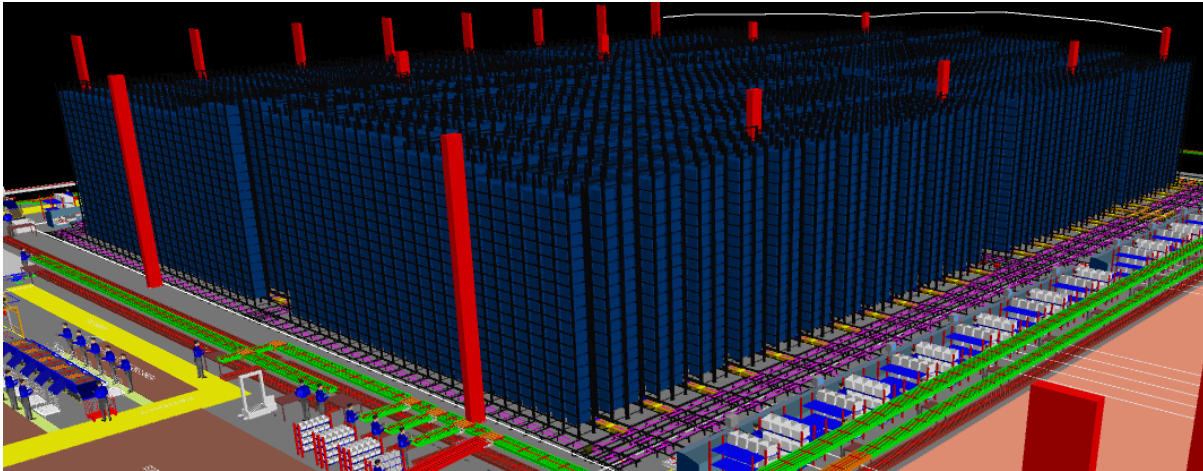
Note : Layout and numbers defined above may change at detailed engineering stage due to incorporation of in-rack sprinklers etc.

7.2.2 Solution Layout Description:



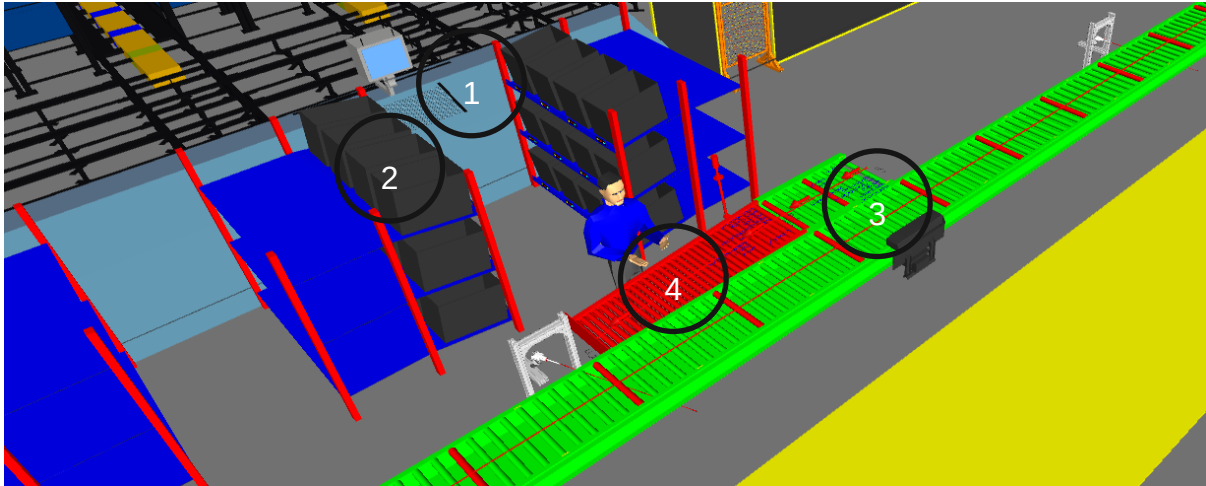
1. Storage Grid – 41812 Bins ; Variable Height : Max 12.05 ; Min 10.05 meters
2. Put-away Stations – 5 Nos with 24 PTL's each. These stations can be used for picking as well
3. Conveyors for handling incoming cartons for put-away .
4. Picking Stations – 11 Nos with 24 PTL's each.
5. Conveyors for handling packed goods from the picking stations to the sorting line and under hauling conveyors from bringing empty boxes to the pick stations
6. Ecommerce consolidation area with PTL set up
7. Ecommerce package sorting Line
8. Store order packed boxes sorting section

7.2.3 Storage Grid:



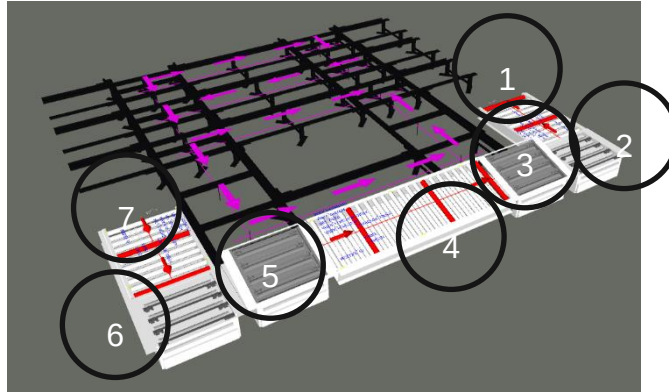
- Maximum Height – 12.055 meter
- Minimum Height – 10.5 meters
- Number of Aisles – 23
- Total Bin Positions – 41812
- Total Storage Capacity – 255118 cubic feet
- Storage capacity considering 90% utilization of bin volume – 229606 cubic feet.
- Total Footprint of storage grid ~ 4200 square meter

7.2.4 Put-Away stations & Conveyors:



1. Put away GTP with HMI and scanner (RFID)
2. PTL's – Fungible for Put-away as well as Picking operations
3. Diverts for selective box diversions to designated stations
4. 4 Gravity accumulation conveyor for accumulation inbound boxes

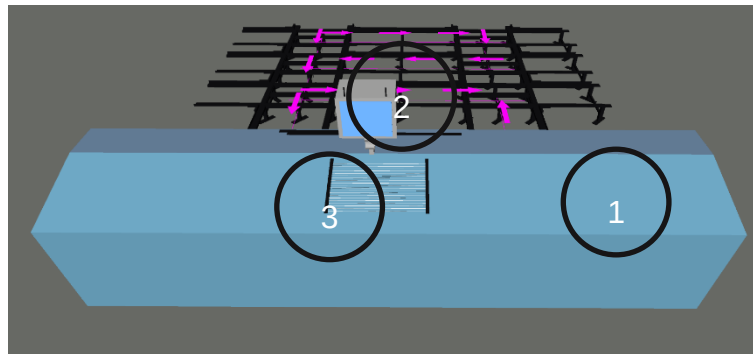
7.2.5 Picking Stations & Conveyors



1. Receiving module – Bots will drop the bins at this position.
2. Pop-up module – This section will receive the bins and will transfer at 90° to the right
3. This is a tilter module which will receive the bins at 0° or flat condition then will tilt to α° and transfer to the next module.
4. This is the pick/put and buffer module which will take the bins to the pick/put window . This is a permanently tilted module for ease of picking. At any point of time this module can store 3 bins ,i.e. 1 bin for picking and 2 waiting in the buffer.
5. This is a rev-tilting module which will receive then bins once the picking or put-away is done in tilted condition , then tilt back to 0° to transfer to the exit pop-up module.
6. Exit pop-up module. This module will also be connected to the next stations. The algorithm is such than if the bins from this station is required to the immediate next station it will get transferred through this module and a bot transfer will not be required.
7. This is the exit module from where the bot will pick up the bins.

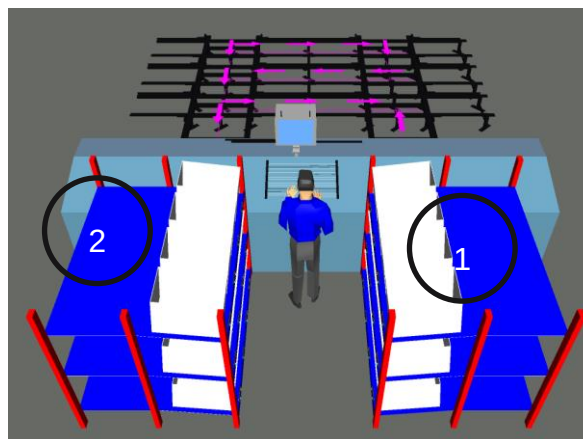
Note: At any given point of time this design can buffer 3 bins + 1 from which the picking is happening thereby reducing the station idle time as much as possible.

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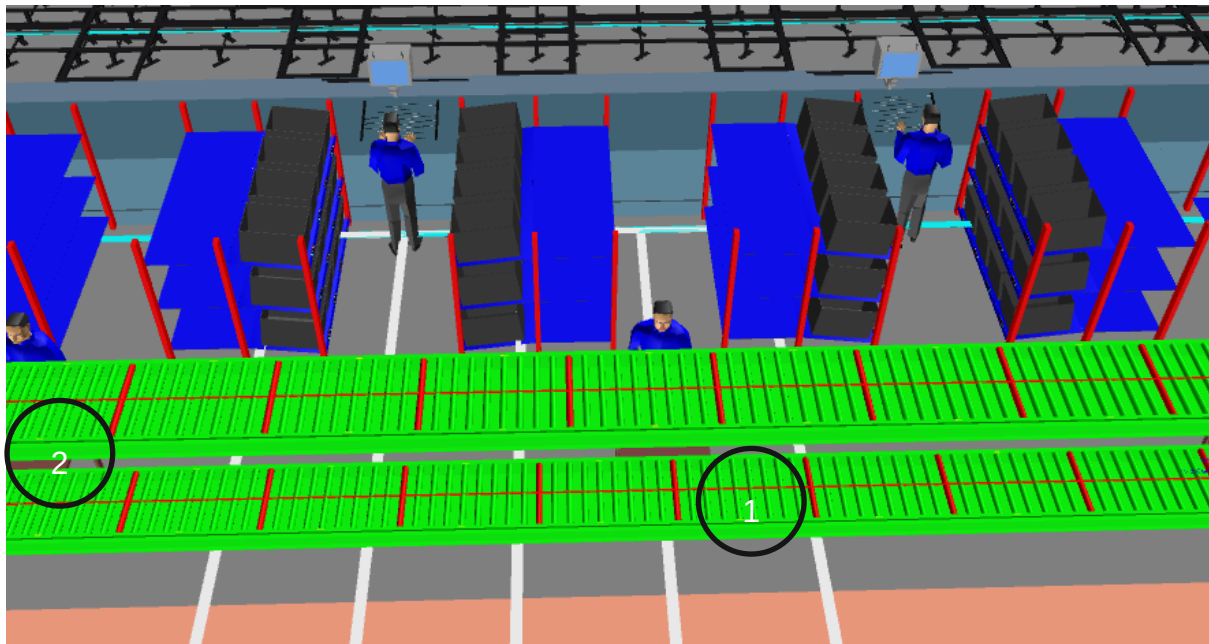
1. GTP Station canopy
2. GTP station HMI with scanner.
3. Pick/put window with light curtain and segment highlighter.

Note: We will be integrating RFID scanner once Decathlon shares the RFID mapping data with Falcon team.



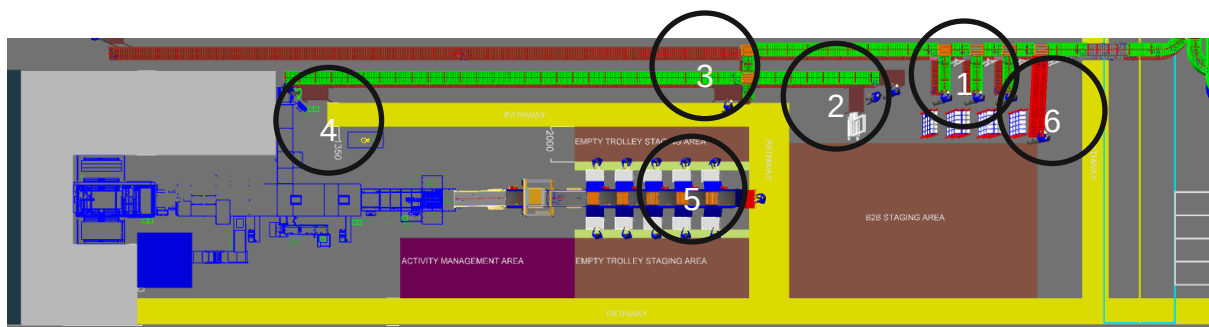
1. PTL Racks – 24 Lights in each station
2. Push Back rack – To push back completed bins for further processing like packing , label printing/pasting etc.

Note: Each PTL racks will have utility trays for printer etc.



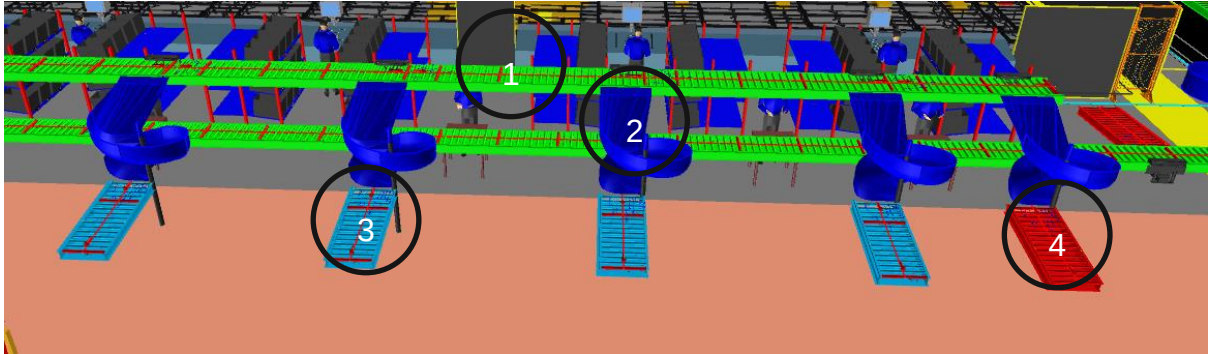
1. FG Conveyor – Packed boxes (for stores, B2B etc.) and picked bins (for Ecom etc.) will be kept on this conveyor
2. Packing tables for store orders, B2B etc.

7.2.6 Ecommerce Consolidation Area



1. Multi order consolidation area – 3 PTL zones with 40 PTL's at each zone.
2. Multi-order folding/banding and label pasting stations
3. Single/Mono order folding and label application station
4. Auto-packer feeding area
5. Auto-packer output sorting for courier partners
6. B2B , Non Standard & Bike orders sorting and staging area

7.2.7 Store Order Box Sorting Line



1. Overhead conveyors line for store order boxes with side barcode scanner
2. Sorting gates with spiral chutes
3. Gravity accumulation conveyors.
4. Rejection line

8. Process Flow

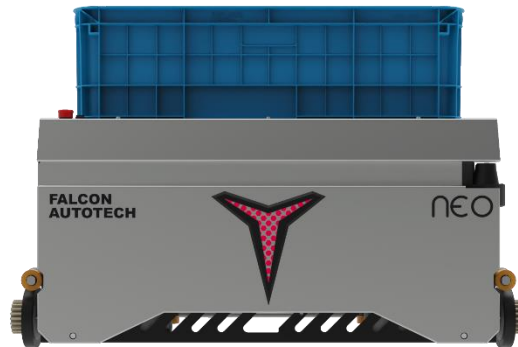
Note: Final process flow to be defined and agreed by FAPL & DSI at the time of BRD

9. Bot & Station Calculations

Outbound Station & Bot Calculation – For the Consolidated Scope			
SI No	Description	Value -14 Hrs Ops	Remark
1	Total Outbound in Eaches	110000	Input
2	Total No of Orders	1183	Case
	No of Eaches/order	30	Input
3	Total Order Lines	35484	Case
4	Hours of operation	14	Input
5	No of Eaches picked/hr	7857	Case
6	Eaches/Order line	3.1	Input
7	Average Eaches/Picked par Bin presentation	4.8	Input
8	No of order lines/hr	2535	Case
9	Total No of Bin presentation	23012.6	Case
10	Station Productivity	650	Input
11	No of Stations Required	12	
12	No of Bots Required	72	
13	Bin Presentation /hr	1644	
14	Bin Presentation/hr / bot	22.7*	Input
Inbound Station & Bot Calculation – For the consolidated Scope			
1	Total Inbound Eaches	110000	Input
2	Put-away Productivity/station/hr	2500	Input
3	Hours of Operation	9	Input
4	No of stations required	4.9	
5	Bin presentation/station/hr	62	
6	Bin Presentation/hr / bot	22.7	Input
7	No of Bots required/station	2.73	
4	No of bots required for Put-away	13.35	
Investment			
1	Total No of Stations Required	16.98	
2	No of stations considered	16	
3	Total No of Bots Required	86	
4	No of Bots considered	85	Sufficient Buffer

- Case – Represents the specific data set we evaluated and may change for different data sets.
- Inputs – Represents data which are either Decathlon/FAPL mandated or deductions based on other calculations.
- (*) – No of bin presentation/hr is a measure of how many times a bot present different or same bin in different stations and not the number of bin cycles,i.e. complete cycle of a bin coming out and returned to the grid.
- Acceptance criteria will be number of bin presentations per hour in a controlled environment defined and agreed by both FAPL and DSI at the DAP stage.

9.1 NEO



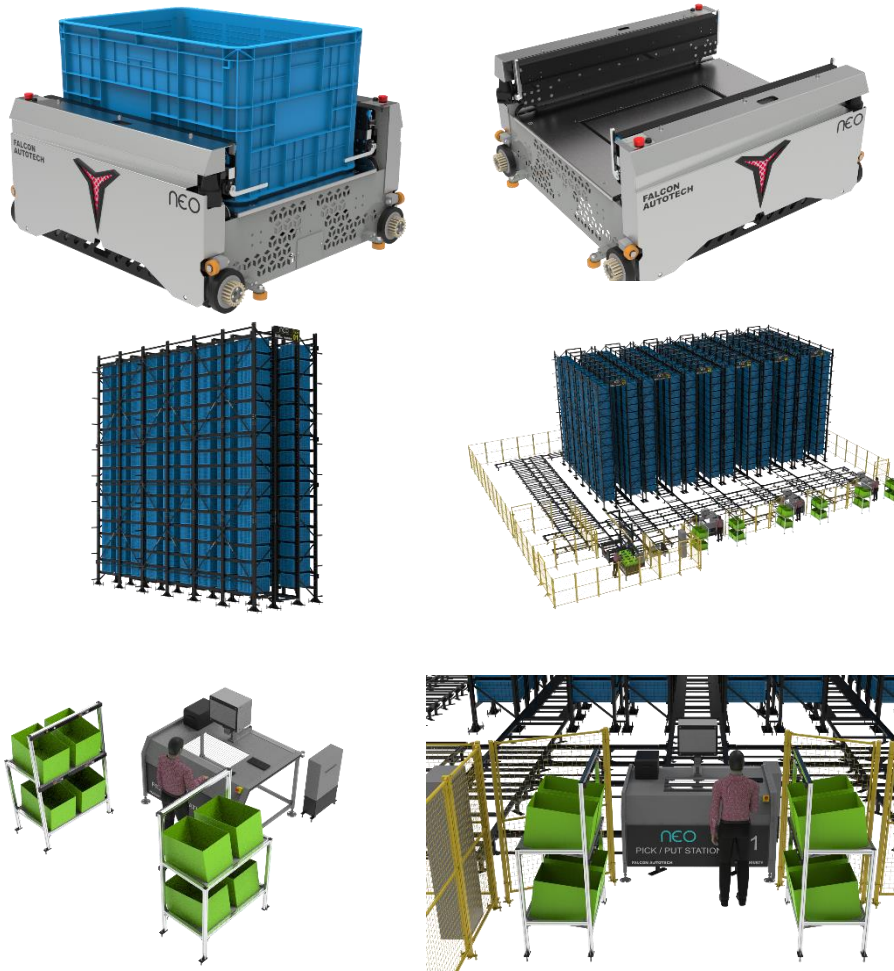
NEO is an end-to-end engineered robotic system for storage and order picking. it is an integrated system consisting of storage bins or totes, storage grids, robots, goods-to-person workstations, and software.

NEO provides higher storage density, is more adaptive, and can be deployed quickly in a cost-effective manner compared to traditional material handling systems. NEO moves flexibly within the storage grid using lidar and AI software capabilities to store and retrieve totes.

- NEO operates on all three axes within the grid to store totes at predetermined locations.
- When the wave starts NEO retrieves and brings the totes having desired products to the picking stations.
- Humans at the picking stations pick items into the order totes. Once the picking finishes NEO stores the inventory totes back to the grid.
- Post completion of orders the order totes can be routed to the packing stations using a series of conveyors.

Technical Specifications

Self Weight	Dimensions	Working Parameters	Performance	Movement	Safety	Fully customizable
200 Kgs Rated load without bin -40 kgs	BOT 1016*906*500mm External Bin Size 810*568*425 mm Bin volume – 5.8 cubic feet	0 to 35 Degrees Navigation mode – QR code	Works 24/7 manual battery swapping	Moves in 3 dimensions X, Y, and Z Maximum operating speed of 2m/s	360 degrees Obstacle detection Lidar in X, Y and Z axis (3D)	Bin Compartment – 2/3/4/6/8 Maximum storage height -12 m Max bins storage per tower -22



NEO Bots: The NEO Bots autonomously and independently move in all three dimensions under the structure and bring products to and from human or robotic picking operators. The robots move at 2 m/s and can carry bins weighing up to 40Kgs. NEO Bots are equipped with a swappable lithium battery which helps the system provide maximum uptime.

Racks: The NEO system is built around a storage grid made up of specialized racking structures that can reach a height of 60 feet to enable vertical space utilization in the warehouse. With zero sensors and electrical cables, the racking structure requires no maintenance.

Bins: Products are stored and moved in specialized bins or totes designed to be handled by NEO system. Each bin can be partitioned into 2/3/4/6/8 sub-sections for better product segregation.

Picking workstations: NEO system comes with operator workstations and PTLs for respective use cases or order picking and storage. The PTL station supports 8/12/16/24 parallel orders by maximizing the commonality of SKUs.

NEOIT: NEOIT is Falcon's in-house developed warehouse control software (WCS) that controls the NEO System. NEOIT does task planning, route planning and records the real-time positions of bins and NEO Bots.

TECHNICAL SPECIFICATIONS		
Basic Specifications	Specification Parameter	Parameter Description
	Model Designation	NEO
	Rated Load	40 Kg
	Self-Weight	200 Kg
	Position Accuracy	+ - 2 mm
	Dimensions (L*W*H)	1016 * 906 * 500 (mm)
	Ground Clearance	20 mm
	Navigation Mode	QR CODE
	External bin Size (L*W*H)	810 * 568 * 425 (mm)
	Bin Volume	5.75 Cubic Feet
	Bin Compartment	2 / 3 / 4 / 6
	Maximum Storage Height	14 Metres
	Maximum Bin Storage per tower	22 Levels
Performance	Driving Mode	Electric Servo Motors
	Control Mode	Manual / Automatic
	Movement	3-D (X, Y, Z Directions)
	Motion Signal	Light Prompt
	Works	24/7
Battery Specifications	Driving Voltage	DC48 V
	Battery Life	Upto 2500 Cycles @ 1C DOD 80%
	Endurance	6 to 8 Hours
	Charging time	Upto 2 Hours
	Charging Mode	Manual Battery Swapping
	Obstacle Detection	360 Degrees (LIDAR in X, Y, Z Axis)
	Emergency Stop	Front and rear emergency stop buttons
	Communication	Industrial Wi-Fi IEEE 802.11 a/b/g/n
	Pick / Put (PP) Station Safety	Light Curtain Sensors
	Working Area Safety	Doors equipped with Saftey Locks
	Safety on Z - Axis Movement	Self Locking Brakes
	Global Saftey (for all Robots)	Using Global Emergency Stop at PP Stations
Operating Parameter	Max Operating Speed (With load) in X & Y Axis	2 m/s
	Max Operating Speed (With load) in Z Axis	1 m/s
	Stopping angle Accuracy	+ - 0.5°
	Stopping position Accuracy	+ - 2 mm
Special Features	Pick / Put Guidance	Yes, through focus light
	Low voltage Battery alarm	Yes, robot will go to the charging station automatically in case of Low Voltage
	Pick / Put Stations Interchangeability	Yes, any station can be used as Pick / Put Station
	Handling fast moving, non-fast moving SKU's to locations near to stations	The system Dyanmically stores the fast- moving SKU's closer to the pick stations and the slow movers farther inside the grid. This enables shorter trip time of the bots thereby greatly reducing the number of robots required.
	FIFO option for the SKU's	It enables the user to consume the SKU which were replenished basis their PUT timestamp.
	Bin recall Feature	It helps in audititng the SKU's
	Re-put Away Feature	It consolidates the less quantity SKU's which are kept in different locations in a single bin and frees the space for next SKU replenishment.
	Realtime Inventory Tracking	Real Time Inventory on the Dashboard
	Ease of maintenance	Mobile Tablet based control for maintenance
	Data Analytics	Detailed and Customized data extraction is possible
	Pick / Put Assist	Dashboards at each PP station that assists operator in pick / put to reduce to reduce the chances of error on the station.

9.2 Powered Roller Conveyor

Falcon's Roller conveyors are highly energy efficient and require negligible maintenance leading to very minimal Operating costs and thus lower cost of ownership.

Some Salient Features of Falcon's roller conveyor:

- Low Noise
- Maximum uptime.
- Minimal maintenance
- High safety standards.
- Fastest ROI
- Comes with MS frame.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Material of Roller	Type	MS Rollers with Zinc Plating
Overall Width	mm	600 or as par Layout
Conveyor Height	mm	Variable as per design shared
Drive Power Rating	Watt	40
Type of Motor	Type	DC Rollers (MDR)
Ingress Protection	Type	IP 54
Type of Drive System	Type	Poly V Belt Drive
Load capacity/unit length of Conveyor	kg/m	30
Roller Span	mm	Variable
Roller Diameter	mm	50
Pitch of Rollers	mm	100
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

9.3 Pop- Up Unit

Falcon's Belt Popup device is used for 90° transfer of products.

Some Silent Features of Falcon's Pop-Up:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Carton Box cross-transfers (Pop-Up Units)	Nos	6
Manufacturer Name	Name	Falcon/Pulse/Itoh Denki
Dimensions	mm	Suitable Products (600 x 400 x 400 MM)
Type – Equipment	Type	Stand Alone
Type – Operation	Type	Yes
Mechanism for Pop-Up	Type	Electric
Drive Power Rating (Lifting)	Watt	40
Type of Motor	Type	DC
Drive Power Rating (For Rollers)	Watt	40
Type of Motor	Type	DC
Ingress Protection type	Type	IP 54
Provision for Carton Box / Carton Detection Sensors	Yes/No	Yes
Load Capacity of Carton Box	Kg	25
Sorting Capacity	PPH	1200

9.4 Swivel Wheel Diverts

Falcon's Swedi device is used for 30°/60°/90° transfer of products.

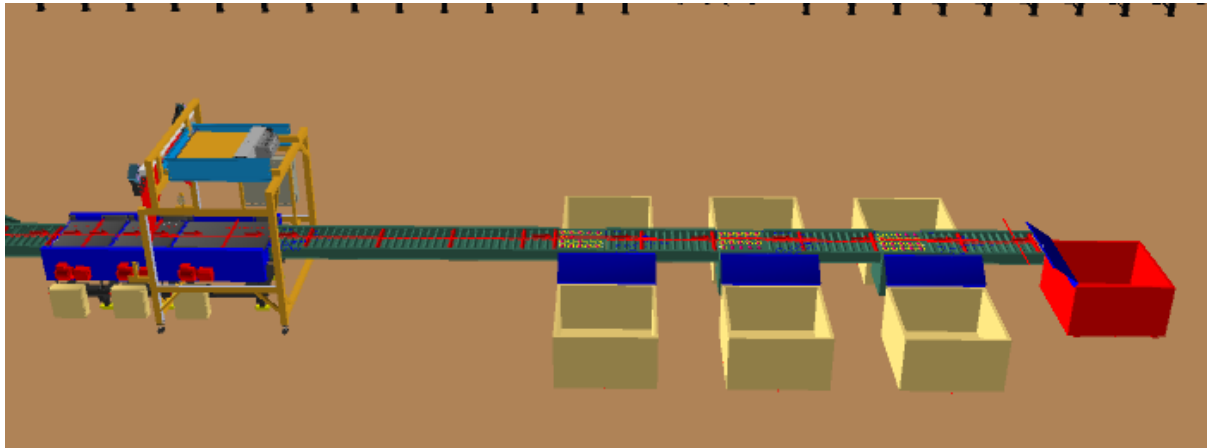
Some Silent Features of Falcon's Swedi Unit:

- Low Noise
- Maximum uptime.
- Electric based popup.
- Minimal maintenance
- High safety standards.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Suitable Carton Size	MM	Min: 250 x 100 x 50 Max: 600 x 400 x 400
Suitable Carton Weight	Kg	Min 5, Max 50
Swedi Divert Units	Nos	1
Dimensions	mm	As per layout
Type - Equipment	Type	Merge/Divert
Type - Operation	Type	Divert
Mechanism for Swivel	Type	Electric servo motor
Roller drive	kW	Powered smart wheel
Roller pitch (L x W)	Type	110 x 120 mm
No. of divert sections	Yes/No	1
Max. product weight (kg/m2)	kg/m2	50
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

9.5 Pop-up Sorter for Ecommerce Parcels



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Suitable Carton Size	MM	Min: 250 x 100 x 50 Max: 600 x 400 x 400
Suitable Carton Weight	Kg	Min 200 gm, Max 10 Kg
Pop-up Sorting Units	Nos	5
Sorting Destinations	Nos	10
Sort fail/Rejection	Nos	1
Throughput	PPH	1200
Dimensions	mm	As par Layout
Type - Equipment	Type	Merge/Sorting
Type - Operation	Type	Sorting
Mechanism for Pop-Up	Type	Electric motor
Drive	kW	AC Geared Motor
Max. product weight (kg/m2)	kg/m2	10
Type of Mounts	Type	Fixed Heights Legs with required Grouting Provisions

9.6 Pick/Put to Light Module

PTL system is falcons in house developed modular light hardware modules and a connected software platform to ensure a flexible, robust, seamless, and easy to setup PTL solution which is fully integrated with Client's WMS/ERP Systems.

The Software platform is powered by a deep artificial intelligence engine that continually mines the data generated by operations and keeps on altering the PTL configuration to increase the system productivity and give you an extra competitive edge.

The System involves a collection of physical sorting locations representing either a customer order/ address location or any other logical sorting group. Each physical location is mapped to a light module which glows on the event of scan of a product for that sorting location.

It is a paperless technology that provide high throughput with single operator with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Falcon Autotech
Dimensions	Mm	Around 165 X 50
PTL location	Nos	As per Layout requirement
Weight of the carton box	Kg	25
LED Display	Type	3 Digit Alpha numeric Rotating Display
LED Size	Mm	~ 25 X 20
LED Indicator	yes/no	YES
Acknowledgment Button	yes/no	YES
Function keys	yes/no	4 Keys
Power Requirements	Type	3 Phase UPS Power
Communication	Type	TCP/IP and Serial
Operating temperature	Deg.	45 Degrees
Software Interface	Type	Over APIs or other methods
Type of mount	Type	Clip On Type on Rails

9.7 Automatic Barcode Scanners

The system is integrated with automated inline barcodes scanning capable for scanning 1D barcodes with high accuracy.



Specification	UOM	Remark
Manufacturer Name	Name	Sick/Cognex
Barcode Type	Type	1D/2D
Module Size	Mils	≥ 10 mil
Max Product Size	Mm	Min: 250 x 100 x 50 Max: 600 x 400 x 400
Number of Codes	Nos	1
Location	Side	Top Side scanning
Orientation	Type	Unidirectional
Color of bars	Color	Black
Under Foil	yes/no	No
Code Length	Mm	To be decided
Code Height	Mm	To be decided

9.8 Hand-Held Barcode Scanners

The system is equipped with handheld barcode scanners capable of scanning 1D barcodes.



9.9 NEO Bins –

Specification	UOM	Remark
MOC	NA	Plastic
Type	Type	Plastic Molded
Dimensions	mm	805 x 550 x 405
Bin CFT	CFT	5.95
Empty Weight	Kg	

10. Proposed System Technical Details/BOM

10.1 Mechanical equipment

Pos.	Qty.	Description	Value
FM1	1	NEO Automation Setup	
		Neo Racking and Structure	42,812 Bin Locations
		Neo Pick/Put Stations	16 Nos
		Neo Bots	85 Nos
		Infra for Neo	1 Set
		PTL Push Back Racks (4X3 Type)	32 Nos
		PTL Modules with Ducts	384 Nos
		PTL Control Boxes	28 Nos
		Hand-Held Barcode Scanners	16 Nos
FM2	1	Conveyor and Sortation Automation	
		Powered Roller Conveyor	320 Meter
		PVC Conveyor	6 Meter
		Idler Roller Conveyor	24.5 Meter
		Angle Merge	01 Nos
		60° Roller Turn	1 Nos
		90° Roller Turn	3 Nos
		Auto Spiral Conveyors	2 Nos
		Divert Units	24 Nos
		Scanning Conveyor	1 Nos
		Side Barcode Scanners	8 Nos
		Auto Barcode Scanning Tunnel	1 Nos
		Hand-Held Barcode Scanners	3 Nos
		Swedi Units	1 Nos
		Packing Tables (All Areas)	Client Scope
		Gravity Spiral	1 Nos

10.1.1 Conveyor BOM

S No.	Name	Conveyor Length (m)	Conveyor width (mm)	Set
12	Gravity roller conveyor	2	600	1
13	Gravity roller conveyor	2	600	1
14	Gravity roller conveyor	2	600	1
15	Gravity roller conveyor	2	600	1
16	Gravity roller conveyor	2	600	1
18	90° Turn Roller			1
19	90° Turn Roller			1
20	90° Turn Roller			1
21	Angle Merge			1
22	60° Turn Roller			1
23	S3 Belt conveyor	3	810	1
24	S3 Belt conveyor	2	810	1
25	Scanning Conveyor	1.2	810	1
26	S3 Belt conveyor	1	810	5
27	Straight Roller Conveyor	19.75	600	1
28	Straight Roller Conveyor	22.625	600	1
31	Straight Roller Conveyor (Accumulation) (600mm)	10.25	600	1
35	Straight Roller Conveyor	1.125	600	1
37	Straight Roller Conveyor (Accumulation) (600mm)	5.5	600	1
38	Straight Roller Conveyor (Accumulation) (600mm)	2	500	1
40	Straight Roller Conveyor (Accumulation) (600mm)	0.9	500	1
45	Straight Roller Conveyor	25.6	600	1
47	Straight Roller Conveyor (Accumulation) (600mm)	1.1	600	1
50	Straight Roller Conveyor	6.25	600	1
53	Straight Roller Conveyor	7.625	600	1
56	Straight Roller Conveyor	4.875	600	1
57	Straight Roller Conveyor	4.875	600	1
58	Straight Roller Conveyor	4.875	600	1
59	Gravity_roller_conveyor	2	700	1
60	Straight Roller Conveyor	5.375	600	1
61	Straight Roller Conveyor (Accumulation) (600mm)	1.375	600	1
62	Gravity_roller_conveyor	2	700	1
63	Straight Roller Conveyor (Accumulation) (600mm)	1.5	500	1
64	Straight Roller Conveyor (Accumulation) (800mm)	3.625	700	1
65	Gravity roller conveyor	2	700	1
66	Straight Roller Conveyor (Accumulation) (800mm)	2.625	700	1
67	Straight Roller Conveyor (Accumulation) (800mm)	3.625	700	1
68	Straight Roller Conveyor (Accumulation) (600mm)	0.625	600	1

69	Gravity roller conveyor	2	700	1
70	Straight Roller Conveyor (Accumulation) (600mm)	1.375	600	1
71	Straight Roller Conveyor	68.75	600	1
72	Straight Roller Conveyor (Accumulation) (800mm)	3.625	700	1
77	Straight Roller Conveyor	1.125	600	1
78	Straight Roller Conveyor	1.125	600	1
79	Straight Roller Conveyor	1.125	600	1
80	Straight Roller Conveyor	1.125	600	1
81	Straight Roller Conveyor	1.125	600	1
82	Straight Roller Conveyor (Accumulation) (600mm)	1.125	600	1
84	Gravity roller conveyor	4.5	700	1
86	Gravity roller conveyor	2	700	1
87	Straight Roller Conveyor (Accumulation) (600mm)	2	500	1
88	Straight Roller Conveyor (Accumulation) (600mm)	1.1	600	1
89	Straight Roller Conveyor (Accumulation) (600mm)	2	500	1
90	Straight Roller Conveyor (Accumulation) (600mm)	1.1	600	1
93	Straight Roller Conveyor (Accumulation) (600mm)	0.9	500	1
96	Straight Roller Conveyor	1.125	600	1
97	Straight Roller Conveyor	54.25	700	1
101	Straight Roller Conveyor (Accumulation) (600mm)	5.125	600	1
102	Straight Roller Conveyor (Accumulation) (600mm)	6.25	600	1
103	Straight Roller Conveyor (Accumulation) (600mm)	1.875	600	1
104	Straight Roller Conveyor	20	600	1
105	Straight Roller Conveyor (Accumulation) (600mm)	7	600	1

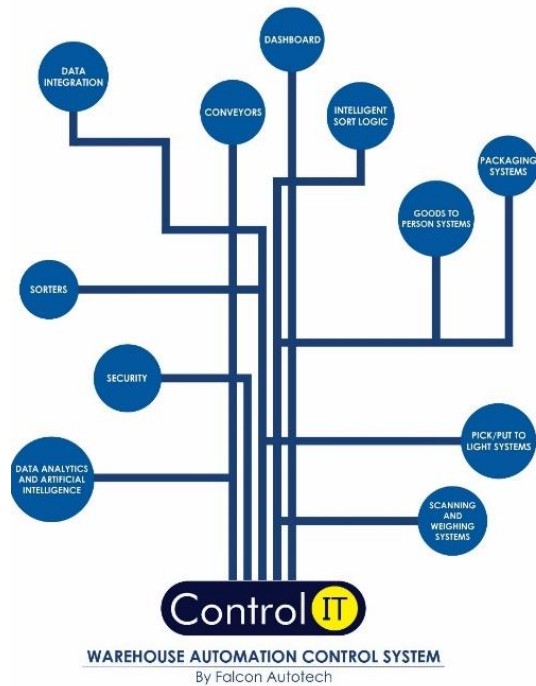
10.2 Electrical Equipment

Electricals			
FC1	1	Electrical Cabling and Wiring with Rated Power	
		Main Power Distribution panel	
		Main Control Panel	Included
		Network Switches	
		Field Cabling	
		GTP Station Rated Power (Total)	20 KW (UPS)
		Battery Charging station Rated Power	48KW (non-UPS)
		Conveyor System	120 KW (UPS)
		<i>*Power ratings are estimates, for capacity building, and may change at the DAP stage</i>	

10.3 Control System

Components			
FC2	1	Controls and IT Infrastructure Controls	
		Barcode Scanner Controls	
		Conveyor Controls	
		Sorter Controls	
		IT	Included
		Control IT Software Package for Neo System	
		Control IT Software Package for Conveyor System	
		Networking Switches	

11. Falcon's WCS CONTROLIT



Key Values

- Supports Multiple Use Cases**
 - Order Consolidation, SKU wise Sorting, Route Level Sorting, Transporter Wise Sorting and many more
- Supports dynamic and static Sort Logic Definition and unlimited Sort Logic Wave definitions**
- Supports various peripheral Equipment Inputs and Outputs**
 - Automated Scanners, Manifest Printers, Label Applicators and many more
- Ability to define Custom Business Logics**
 - Cut Off Timings, Quantity/Weight and Volume Thresholds etc
- Security**
 - User profiling with Role Right Mapping
 - Database Encryption
 - Secured and Protected Communication between WMS and WCS layers

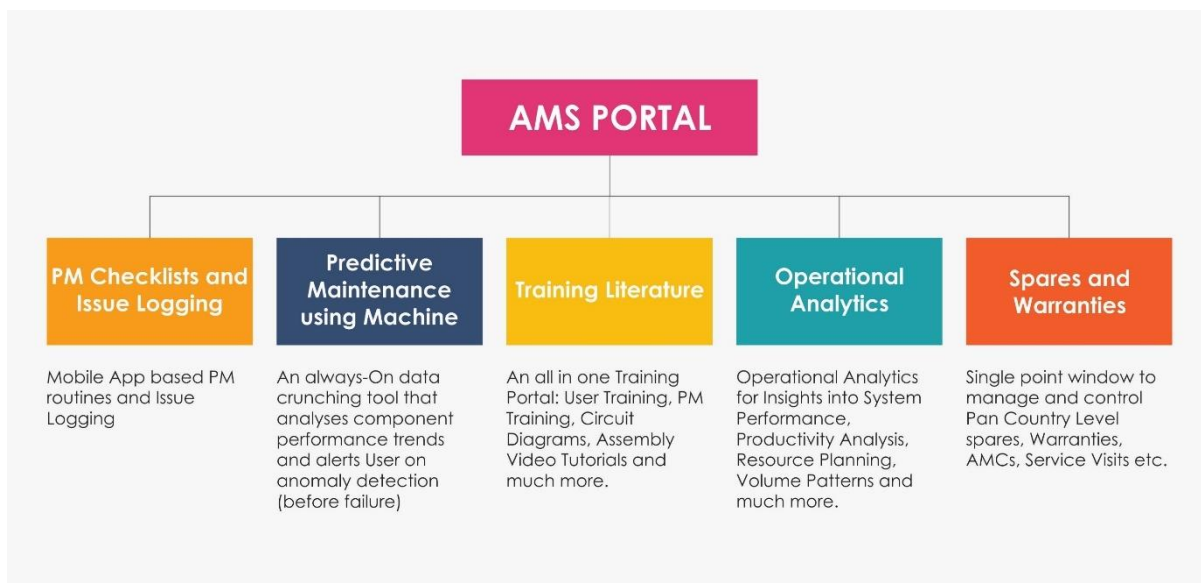
Supported Protocols

API	FTP	FILE UPLOAD / DASHBOARD	ODBC
WEBSERVICES	AMAZON S3 SERVER	FIREBASE	Any Customer Protocol

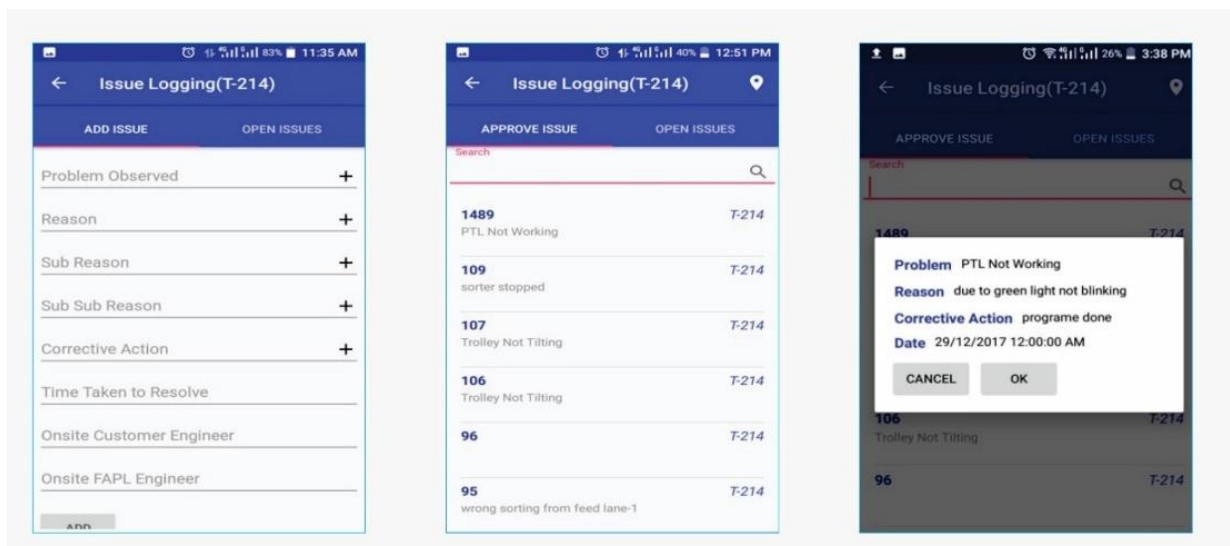
12. Key Components Make

Component	Make
PLC & RIO	Siemens/Omron
VFD	Siemens/Omron/Lenze/AB
Ethernet switch	Siemens/Phoenix Contact
SMPS	Omron/Selec/Meanwell
Line chokes	EMIS/GURU NANAK ELE.
Slim relays	Phoenix contact
Terminals	Wago
Phase failure relay	Selec
Door limit switch	Salzer
Cable lugs	Dowell's/TRINITY TOUCH
Tube light	Philips
Tower lamp + Hooter	Leuze
Utility switch & socket	Anchor
Fan/Air conditioner	Rexnord
Switchgear	Schneider
Main load break switch (CP)	Schneider
Popup Units	Pulse/Itoh Denki/Falcon
Spiral Conveyor	Ambaflex/Apolo/Similar

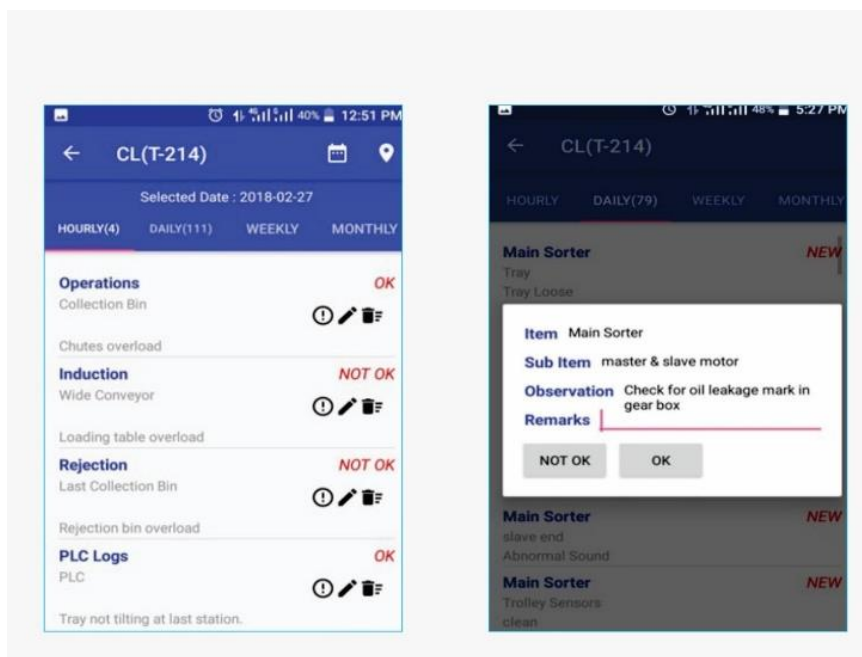
13. Automation Management System (AMS)



13.1 Issue Logging



13.2 Preventive Maintenance Check List



13.3 Predictive Maintenance



13.4 Training Portal

Employee Training Admin

Employee Name: Training Type: Search

Show All Scheduled Completed

Employee Name	Training Type	Subject	Media	Training Value By	Actual Start Date	Actual End Date	Setting
Blank Page (1/1)	Product Training	T8 Tray Sorter T8 Tray Loop Sorter Sorter		Video Checkpoint 1			Download
Blank Page (1/1)	Product Training	BELT CONVEYOR WITH REJECTION AND SORTER		Select			Download
Blank Page (1/1)	Product Training	T8 Tray Sorter SHELVE SORTER Sorter Sorter PTL POPUP SORTER FALL		Select			Download
Blank Page (1/1)	Product Training	VOL Custom Conveyor Check Wagoner Belt Type Check Wagoner Custom Belt Conveyor					Download
Blank Page (1/1)	Product Training	T8 Tray Sorter SHELVE SORTER Sorter Sorter					Download

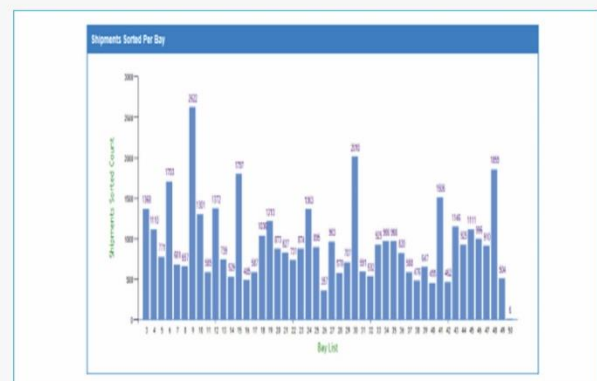
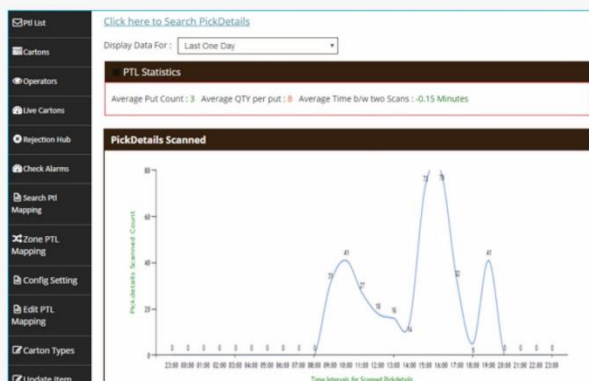
Training Material Master Admin

Training Type: Document Name: Video Name: Search

Select Page Size:

Training Type	Subject	Document Name	Training Document	Video Name	Training Video	Time Limit	Setting
Comprehensive Training	K8 TRAINING	SCNPS MANUAL	download			2	Download
Comprehensive Training	WFO TRAINING	POWERPLEX 400 MANUAL	download			2	Download
Comprehensive Training	PTL	PTL INSTRUCTION MANUAL	download			2	Download
Comprehensive Training	PTL	operational manual	download			2	Download
Product Training	T8 Tray Loop Sorter	T214 Secondary sorting manual	download			2	Download
Maintenance & Troubleshooting	T8 Tray Sorter	T214 MAINTENANCE MANUAL	download			2	Download
Product Training	T8 Tray Loop Sorter	T214 GENERAL MANUAL	download			2	Download
Product Training	T8 Tray Loop Sorter	T214 INDUCTION LINE MANUAL	download			2	Download
Department Procedures	Field & Power Wiring	Electrical-01	download			20 mins	Download

13.5 Operational Analysis



14. Program Organisation

14.1 Program/ Project Schedule

Go Live – 7 months from the date of PO or advance receipt which ever is later.

Note: Final project schedule will be conveyed at the DAP stage.

14.2 Program Management

For this program, proposed approach covers the following aspects:

1. Creation and monitoring of the project plan.
2. Communication with Decathlon
3. Scheduling of the resource management.
4. Management of the risks and opportunities.
5. Management of the list of anomalies or reservations.

15. Commercial Terms

15.1 Price Sheet – Scope Definition and Itemised Cost

Itemized Cost		
1	NEO Racks & Tracks	₹ 12,57,42,101
2	NEO Automation	₹ 21,39,48,745
3	NEO Bins - Plastic without partitions	₹ 4,83,00,000
Total		₹ 38,79,90,846

The Total Price is to be understood.

- Ex Works – Greater Noida, India
- Including all-round wire mesh fencing & safety doors with interlocks.
- Includes walkovers and emergency exits as per layout.
- Freight charges shall be extra at actual.
- Taxes and duties shall be extra as applicable.
- Including Packing charges
- Installation and Commissioning: Included
- Price is valid for 60 days from the date of proposal.

*Material Unloading & Laydown area is in customer scope

15.2 Payment Terms

The proposal is based on the following payment schedule:

Stage 1: 5% Advance along with PO

Stage 2: 5% Against DAP Approval

Stage 3: 10% against BRD Approval.

Stage 4: 40% against material readiness intimation

Stage 5: 20% with 100% taxes against material delivery in pro rata basis

Stage 6: 10% against Installation or 60 days after delivery, whichever is earlier. (If delay is on account of FAPL activity the alternate clause will not be applicable)

Stage 7: 10% against successful commissioning or 90 days after delivery, whichever is earlier (If delay is on account of FAPL activity the alternate clause will not be applicable).

16. Warranty & SLA

Falcons offered System comes with a standard warranty of 1 year, and an additional 1 year as a special case for this project, for complete system against manufacturing defect. Any damage caused to the system by an external source, abnormal use of equipment or any other factor not attributable to Falcon Autotech shall not be covered under warranty. Warranty period will start once the system is live and DSI starts using the system for its day to day operations.

Post completion of warranty, customer can opt for Extended warranty. Warranty covers the following support:

- Telephonic, Email and Remote Service Support when required.
- Regular Software updates and Bug Fixes.
- Supply of Mechanical and Electrical components in case of failure (excluding damages as mentioned in Exclusion Clause)

The warranty does not apply to replacement or repair of:

- Normal wear and tear.
- Consumables.
- Faulty articles:
 - Failure to comply with the manufacturer's recommendations (logistics documentation, Technical Information Note, retrofit document) and the rules of the trade.
 - Negligence or abnormal use of equipment.
 - Anomalies produced by an environment of use, storage or transport that does not comply with the specifications or recommendations of Falcon: packaging, temperature, hygrometry, sector, insulation, etc.
 - A defect due to a cause external to the supplies and services of Falcon.
- Equipment other than that supplied by Falcon.
- Items that can be repaired exclusively by Falcon that have been repaired or attempted repairs other than those carried out by Falcon.
- Items that fail due to normal wear and tear of one or more of its components or whose tamper-evident seals (varnish, strip, etc.) have been broken or whose serial numbers have been removed or modified.
- Items damaged during transport to Falcon due to the use of unsuitable packaging.

16.1 Supply and Service Terms

The supply of material will be done by Falcon Autotech through its manufacturing unit in India. The installation and commissioning of the system will be done by Falcon along with our local partner.

The maintenance of the system during the warranty period will be done by Falcon. Any service visit required during the warranty period due to any manufacturing defect in the system will be done by Falcon. In case a spare part needs replacement while the system is in the warranty period, the same will be supplied and replaced by Falcon FOC. However, Decathlon is advised to keep critical spares parts recommended by Falcon in its own stock so that the same can be used without any delays. During

the warranty as well as if the system is under CAMC post the expiry of the warranty period the same will be replenished by Falcon FOC.

Post warranty Falcon offers various service packages:

- Extended Warranty
- Consumable package
- Service Package
- On-site Support
- Comprehensive AMC

Falcon is offering Decathlon the Comprehensive AMC package which takes care of all the four other packages.

Consolidate Scope	
Item Description	CMC +OSS Expenditure
Year 1 Warranty + OSS	₹ 24,00,000
Year 2 Warranty + OSS	₹ 24,00,000
Year 3 (CAMC)	₹ 1,56,24,903
Year 4 (CAMC)	₹ 1,56,24,903
Year 5 (CAMC)	₹ 1,64,06,148
Year 6 (CAMC)	₹ 1,72,26,456
Year 7 (CAMC)	₹ 1,80,87,778
Year 8 (CAMC)	₹ 1,89,92,167
Total Exp in 8 years	₹ 10,67,62,356

- Includes all spares.
- Includes all consumables.
- Includes Routine maintenance and Breakdown visits.
- 24x7 remote support via hot line
- Onsite Support
 - 1 mechanical & 1 electrical engineer in each shift
 - 4 persons for 2 shifts
 - 6 days a week.

16.2 Uptime Commitment

Falcon commits 98.5% uptime of its supplied system with the following considerations:

1. Complete Loss of throughput shall be deemed as downtime.
2. Any downtime caused due to factors not attributable to Equipment or Staff of Falcon would be excluded.
3. On Site Support (OSS) will be available for all shifts.
4. Spares must be available on site.
5. 2 Hours of Continuous Preventive Maintenance downtime shall be provided daily.
6. Runtime 22 hours/day

Spares List & Cost

P.NO	PART_NAME	SPARE/CONSUMABLE	Spare Qty
401	SERVO MOTOR	SPARE	13
402	X-Z Axis GEARHEAD	SPARE	6
403	BALL BEARING	SPARE	21
404	TIMING BELT	CONSUMABLE	85
405	BALL BEARING	SPARE	31
406	TIMING BELT	SPARE	5
407	THRUST BEARING	SPARE	17
408	BALL BEARING	SPARE	11
409	LEAD SCREW WITH NUT	SPARE	7
410	Slider GEARHEAD	SPARE	2
411	Lifting GEARHEAD	SPARE	2
412	TELESCOPIC GUIDE	SPARE	17
413	DOUBLE SIDE TIMING BELT	SPARE	9
415	BALL BEARING	SPARE	6
416	TIMING BELT	CONSUMABLE	43
417	BELT	SPARE	17
418	SERVO MOTOR	SPARE	6
419	BALL BEARING	SPARE	16
421	ENERGY CHAIN	SPARE	9
422	Y axis GEARHEAD	SPARE	3
423	SERVO MOTOR	SPARE	6
424	WHEEL	CONSUMABLE	306
426	WHEEL	CONSUMABLE	7
427	WHEEL	CONSUMABLE	7
431	ENERGY CHAIN	SPARE	4
434	CABLE TIE MOUNT	SPARE	85
438	LIDAR 2D	SPARE	26
439	CONNECTOR CABLE	SPARE	6
442	GEAR MOTOR	SPARE	11
443	RUBBER_GROMMET		0
444	SHACKLE		0
701	PROXIMITY SENSOR	SPARE	5
706	REFLECTIVE SENSOR	SPARE	9
708	PHOTOELECTRIC_SENSOR	SPARE	5
709	CONNECTOR_CABLE	SPARE	2
710	REFLECTOR	SPARE	17
711	POSITION SCANNER OPTO SWITCH PGV100R	SPARE	5
712	DIFFUSE_SENSOR	SPARE	5
713	Left Pinion	CONSUMABLE	510

714	Right Pinion	CONSUMABLE	510
715	AL Racks	CONSUMABLE	51

S No.	Part_name	Make	Model_no	unit	Spare Qty
1	PLC (NX1P2) CPU UNIT	OMRON	NX1P2-1140DT1	NOS	7
2	NX-SERIES RS-422A/485 COMMUNICATION OPTION BOARD	OMRON	NX1W-CIF12	NOS	7
3	BATTERY_FOR_PLC	OMRON	CP1W-BAT01	NOS	7
4	NX-SERIES COMMUNICATION INTERFACE UNIT (RS-422A/485)	OMRON	NX-CIF105	NOS	7
5	FL WLAN 1011 Wireless Module	PHOENIX CONTACT	2702900	STD	7
6	ANTENNA CABLE - RAD-PIG-EF316-N-RSMA	PHOENIX CONTACT	2701402 (HSN CODE-85442090)	STD	7
7	Antina ANT-OMNI-2459-04	PHOENIX CONTACT	1284780	STD	7
8	AGV Front Star LED Light	EAGLE WEIGHING	FAL-LED 001-V0	NOS	7
9	Illuminated Emergency Stop Switch with indication Dia 16 mm 24V DC	IDEC	XA1E-LV311Q4R	NOS	7
10	16MM A6 SERIES SELECTOR SWITCHES 90?-2-POSITION	IDEC	AS6M-2Y2	NOS	7
11	8 PIN RELAY	OMRON	MY2N 24VDC	NOS	13
12	8 PIN RELAY BASE	OMRON	PYF08A-N	NOS	9
13	14 PIN RELAY	OMRON	MY4N 24VDC	NOS	13
14	HOLDING CLIP FOR MY SERIES RELAY	OMRON	PYC-A1	NOS	9
15	14 PIN RELAY BASE	OMRON	PYF14A-N	NOS	9
16	Solid-state relay module	PHOENIX	PLC-OSC- 24DC/ 24DC/ 2	STD	9
17	DIN RAIL	TRINITY TOUCH	TDR3-35/7.5-01 17 00284	STD	0
18	TERMINAL BLOCK (2.5 SQ.MM - BLUE)	WAGO	280-104	STD	13
19	END PLATE (2.5 SQ.MM TERMINAL- GREY)	WAGO	280-301	Nos	0
20	ADJACENT JUMPER (2.5 SQ.MM- GREY)	WAGO	280-402	Nos	0
21	END PLATE (4 SQ.MM TERMINAL- GREY)	WAGO	281-301	Nos	0

22	TERMINAL BLOCK (4 SQ.MM- GREY)	WAGO	281-101	nos	0
23	ADJACENT JUMPER (4 SQ.MM- GREY)	WAGO	281-402	nos	0
24	TERMINAL BLOCK (4 SQ.MM - BLUE)	WAGO	281-104	nos	13
25	TERMINAL BLOCK (2.5 SQ.MM- GREY)	WAGO	280-101	nos	13
26	RJ45 CONNECTOR FOR CAT 6A ETHERNET SHIELDED CABLE	AMPHENOL FCI	AFCM2213T13	NOS	13
27	STRAIN RELIEF BOOT FOR CAT 6A RJ45 CONNECTOR	AMPHENOL FCI	FOR CAT 6A AFNS CONNECTOR	NOS	13
28	ETHERLINE FD P CAT.6 4X2XAWG26	LAPP	2170488	STD	13
29	DC-DC CONVERTER 48V (32V-60V) TO 24V 10A	SCIENCE	SG-48S2410	NOS	9
30	DC-DC CONVERTER 48V (30V-60V) TO 5V 10A	SCIENCE	SG-48S0510	NOS	9
31	LFP BATTERY	YUKINOVA	LFP15S 4P15AH	STD	0
32	1 CORE 0.5 SQ.MM BLUE WIRE	LAPP	4510021U	STD	0
33	1 CORE 0.5 SQ.MM GREY WIRE	LAPP	4510061U	STD	0
34	1 CORE 4 SQ.MM RED WIRE	LAPP	4520043U	STD	0
35	1 CORE 4 SQ.MM Black WIRE	LAPP	4520013U	STD	0
36	1 CORE 2.5 SQ.MM BLACK WIRE	LAPP	4520012U	STD	0
37	1 CORE 1 SQ.MM BLUE WIRE	LAPP	4510023U	STD	0
38	1 CORE 1 SQ.MM GREY WIRE	LAPP	4510063U	STD	0
39	1 CORE 2.5 SQ.MM RED WIRE	LAPP	4520042U	STD	0
40	PIN TYPE THIMBLE (0.75 SQ.MM, GREY)	TRINITY TOUCH	F-E7508 GY	STD	0
41	PIN TYPE THIMBLE (1 SQ.MM, RED)	TRINITY TOUCH	F-E1008 RD	STD	0
42	PIN TYPE THIMBLE (2.5 SQ.MM, GREY)	TRINITY TOUCH	F-E2508 GY	STD	0
43	PIN TYPE THIMBLE (4 SQ.MM, GREY)	TRINITY TOUCH	F-E4009 GY	STD	0
44	U TYPE THIMBLE (1.5 SQ.MM, RED)	TRINITY TOUCH	F-SVS1-4	STD	0
45	U TYPE THIMBLE (2.5 SQ.MM)	TRINITY TOUCH	F-SV2-3-2	STD	0

46	U TYPE THIMBLE (4.0 SQ.MM)	TRINITY TOUCH	F-SV5-3.7	STD	0
47	CABLE TIE BLACK (100MM)	JIGO	(100mm)x(2.5mm)	PKT	0
48	CONTINUOUS PLUG IN BRIDGE - FBST 500-PLC GY	PHOENIX	2966838	STD	0
49	CABLE TIE BLACK (300MM)	JIGO	(300mm)x(4.8mm)	NOS	0
50	MCB (2 POLE 40A DC)	SCHNEIDER	A9N61537	NOS	7
51	CABLE TIE BLACK (200MM)	JIGO	(200mm)x(3.6mm)	Nos	0
52	WAGO TERMINAL NUMBERING (1-10)	WAGO	209-502	nos	0
53	4 CORE 0.75 SQ.MM TWISTED CONTROL CABLE	IGUS	CF77.UL.07.04.D	Nos	0
54	WAGO TERMINAL NUMBERING (11-20)	WAGO	209-503	nos	0
55	CABLE DUCT SLOTTED	TRINITY TOUCH	TBWDN40X40LG1	STD	0
56	WAGO TERMINAL NUMBERING (21-30)	WAGO	209-504	nos	0
57	CABLE DUCT SLOTTED (25X40, LIGHT GREY, 2MTR)	TRINITY TOUCH	TBWDN25X40LG2	STD	0
58	WAGO TERMINAL NUMBERING (31-40)	WAGO	209-505	nos	0
59	16 CORE 0.25 MM.SQ	LAPP	34416	MTR	0
60	WAGO TERMINAL NUMBERING (41-50)	WAGO	209-506	Nos	0
61	6PIN BOX MOUNTING RECEPTACLE FEMALE WITH DUST CAP	ALLIED CONNECTORS	MS 3102R 20-8S DC SQG	NOS	0
62	WAGO TERMINAL NUMBERING (51-100)	WAGO	209-507	Nos	0
63	1 CORE 0.25 SQ.MM BLUE WIRE	LAPP	4126002S	STD	0
64	1 CORE 0.25 SQ.MM GREY WIRE	LAPP	4126106S	STD	0
65	Diode	FAIRCHILD	1N4007	Nos	13
66	HEAT SHRINK TUBE 3.5MM (White)	WOER	NA	Nos	0
67	FERULE TUBE 3.2	MARTIX	FERULE TUBE 3.2	Nos	0
68	FERULE TUBE 5MM	MATRIX	FERULE TUBE 5MM	Nos	0
69	SPIRAL SLEEVE(DIA-3MM)	Keshvas	3MM	MTR	0
70	SPIRAL SLEEVE(DIA-9MM)	Keshvas	9MM	Mtr	0
71	END STOPPER	WAGO	249-116	Nos	0
72	YELLOW COLOR LABEL TAPE	MAX	LM-TP512Y	Nos	0
73	FLUX CORED SOLDER WIRE (50 GMS)	BHARTI	RM0748 GRADE-60-40 SEG-18	NOS	0
74	BATTERY CONNECTOR	ICIL	BS4W2PG0 GXF	STD	9
75	RELAY	WAGO	788-313	STD	9

76	ANTENNA CABLE	PHOENIX	2903266	STD	9
77	Panel mounted RJ45 port	SCHNEIDER	XB5PRJ45	STD	9
78	PIN TYPE THIMBLE (10 SQ.MM, RED)	TRINITY TOUCH	F-E10-12 RD	STD	0
79	FERULE TUBE 4.2	MARTIX	FERULE TUBE 4.2	Roll	0
80	ÖLFLEX® 110 FR-LSH 5 G 0.5 1100V GY	LAPP	382100515	STD	0
81	FL SWITCH 1005N	PHOENIX CONTACT	1085039	STD	9
82	FEETECH MOTOR CONNECTOR	NA	NA	STD	9
83	FEETECH MOTOR CONNECTOR CRIMP TERMINAL	NA	NA	STD	9

S.NO	DESCRIPTION	TOTAL COSTING(INR)
1	Total Bot proposed	
2	NEOBOT 1.0 Spare - Mechanical	34,63,605
3	NEOBOT 1.0 Spare - Electrical	10,44,126
4	NEOBOT 1.0 Spare - Field	1,00,000
5	Conveyor Spares (List will be shared post DAP)	35,35,878
TOTAL COST		INR 81,43,609

17. Exclusions

The scope of supply includes all parts which are defined in the Supplier's quotation.

All other parts which are not defined in the Supplier's quotation do not belong to the Supplier's scope of supply and are excluded. The following parts are also excluded:

- A. RFID sensors, tunnels and associated conveyors if any.
- B. Printers, Bar code printers wherever required,
- C. Packing Tables and its associated accessories.
- D. Building infrastructure; building structure, doors, fire exits, levelling devices, building extinguisher and fire alarm system, building heating and lighting system.
- E. Electrical power supply and wiring to the main control cabinets.
- F. UPS for Controls and Drives
- G. Electrical Power for Installation
- H. Network Cabling up to the Main Server Rack
- I. Intermediate wiring to parts which are to be supplied by the Purchaser/others.
- J. Emergency/Uninterruptable power supply
- K. Fire-alarm and fire protection devices.
- L. Traffic and route markings (outside FAPL storage grid)
- M. Laydown Area
- N. Ram protection devices
- O. Cat walks, bridges, maintenance aisles and platforms if not mentioned in the BOM.
- P. Assembly tools like forklifts and hoisting machines
- Q. All kind of network incl. Local Area Network (LAN/WLAN), exceeding the scope described in Scope of Supply
- R. Any Kind of Civil work
- S. Any adjustment of the Supplier's scope of supply to local rules and regulations
- T. X-Ray machines
- U. Roller cages/ Pallets
- V. Simulation and 3D animation of the system
- W. Interface with other equipment not specified in this offer.
- X. Provision of facilities for the control room (furniture, air conditioning, heating, etc.).
- Y. The supply and installation of fencing around the different corridors.
- Z. Any item specifically indicated as not forming part of the subject matter of the Seller's supply in the offer documentation.

18. Annexure*

- *Annexure I-Rev: Cost bifurcation of Falcon scope & Rack supplier scope in the Rack & Track part of the Solution.*
- *Annexure II: Cost bifurcation of different components of the proposed CAMC cover.*
- *Annexure III : Detailed Cross Break-up according to family.*

(*): The annexures are attached separately in the email containing this proposal.

